8 Bearing handling

8-1 Precaution

Bearings are an extremely precision mechanical components. Exercise great care for its handling. Followings are precautions for the handling.

1) Keep bearings and surroundings clean

Foreign matters invaded inside of bearings such as dust and dirt have harmful effect in rotation or operation life on the bearings. Take extra precaution to maintain cleanness of bearing, surrounding components, work tools, lubricants, lubrications oil and working environment.

2) Handle bearings carefully

Shocks such as caused by falling bearing may result in damage or impressions on track or rolling elements. They can be a cause of failure so that handle bearings carefully.

3) Use proper tools

Make it sure to use work tools properly for bearing type for assembling and disassembling.

4) Pay attention to rust

Although bearings are applied with anti-rust oil, handling with bare hands may cause generation of rust with perspiration from hands. Exercise care and use rubber gloves or apply mineral oil to hands when handling with bare hands.

8-2 Mounting

Preparation

Bearings should be mounted in clean and dry circumstance. Dirt on mounting tools should be removed prior to mounting work, then verify that dimension accuracy, shaft and housing roughness and geometric accuracy are within designed tolerance.

Packing of bearings should be opened just before start mounting. Fill lubrication grease without washing bearing in the case of grease lubrication. Washing is generally not required for oil lubrication as well. Still it is recommended to thoroughly wash out oil and grease when application demands high accuracy or lubricating performance is degraded by mixing lubricant and anti-rust agents.

Mounting method

1) Press fitting

In mounting bearings from small to medium sizes which don't need large forces , press fittings in room temperature are conducted widely. In this case, use pressing fixture as shown in Figure-12 to apply force evenly at side of bearing and press it in carefully. Applying high viscosity oil on fitting surface during work may reduce friction on the surface.

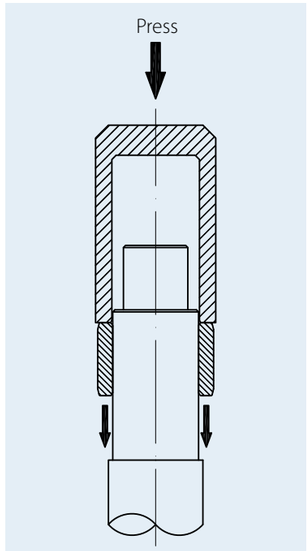


Figure-12 Press fitting of Inner ring

2) Shrink fitting

Shrink fitting is broadly used for tighter interference or mounting large size bearing. How to fit is heating housing for outer ring and inner ring for shaft respectively with pure mineral oil with less corrosivity in order to expand their inner diameter for mounting onto the shaft. Heating temperature must not exceed 120° C. During mounting, inner ring could expand toward shaft direction so that it needs to be pressed against shoulder until completion of cool down to avoid gap between the inner ring and the shoulder.

8-3 Operation inspection

Operation inspection needs to be performed in order to confirm that bearings is properly mounted. Power operation at given speed without operation inspection may result in damage of bearings or heat-seizure due to lubrication failure in the case that mounting is insufficient. Shaft or housing should be rotated by hand after bearing mounting to confirm if there is no abnormality followed by check (or inspection) in stepping increase of speed from no load, low speed operation with power up to loaded operation.

Followings are typical abnormal items and major causes that can be checked in the operation inspection.

1) Check item in operation by hand

- Fluctuation in rotation torque, Insufficient mounting
- Sticks and abnormal noise, Impression, damage, invasion of dirt or foreign matters in track surface
- Excessive torque, Insufficiently small clearance

2) Check item in operation by power

- Abnormal noise, vibration ···· Impression, invasion of dirt or foreign matters in track surface, excessive clearance
- Abnormal temperature ····· Insufficient lubrication, insufficient mounting, insufficiently small clearance

8-4 Removing

Bearing may be removed for periodical machine maintenance or repair for trouble. Bearing and other components should be carefully disassembled in the same manner as the mounting in the case of re-using disassembled bearing or researching trouble condition.

Bearings should be carried out in an appropriate manner in accordance with type of bearing and condition of fits. Structure design should take disassembling work into consideration at planning stage of construction around the bearing since it would be difficult to disassemble especially the tight fit bearing.

Removing outer ring

Installing bolts for disassembling outer ring at several locations in circumference of housing will allow removing outer ring assembled with tight fits easily by tightening-up the screws evenly as shown in Figure-13.

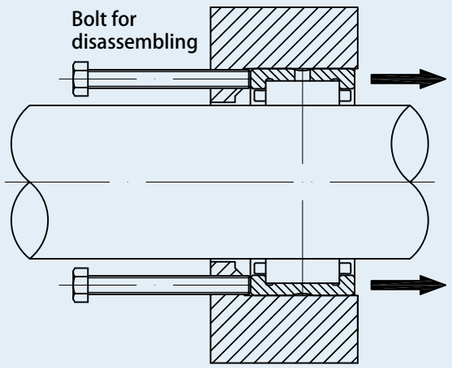


Figure-13 Disassembling outer ring

Disassembling inner ring

Inner rings can be carried out most easily by pulling out by press (Figure-14). Dedicated removal tool (Figure-15) designed in accordance with dimension of the bearing is in use as well.

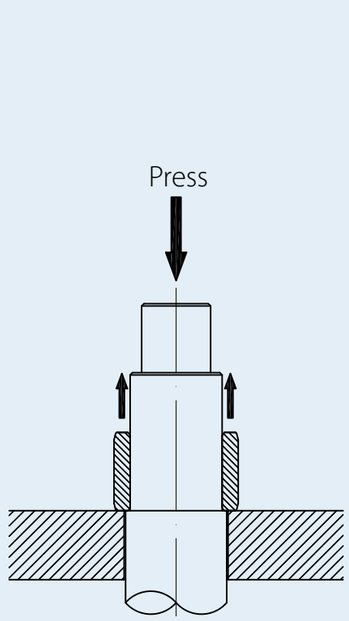


Figure-14 Disassembling inner ring 1

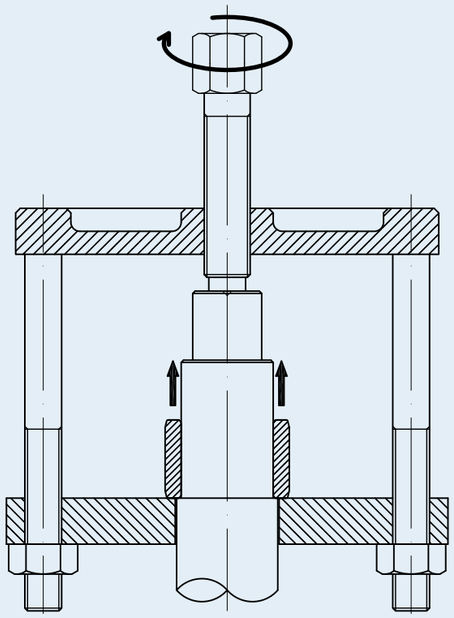


Figure-15 Disassembling inner ring 2

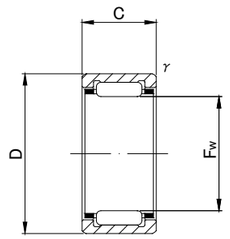
8-5 Maintenance and inspection

Periodical maintenance and inspection are essential for maximizing performance and prolonged usage of bearing as well as early discovery of abnormality of the bearings. Inspection items of bearings under operation include temperature, operation sound, vibration of bearings and condition of lubricant, whose observation enables judging timing of lubricant replenishment and replacement of components.

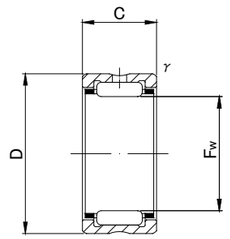
MACHINED RING NEEDLE ROLLER BEARINGS
WITHOUT INNER RING

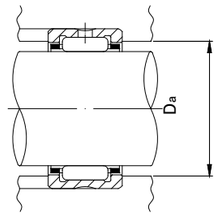






NK(Fw ≦ 10)





RNA49,RNA59,RNA69(Fw ≦ 35),NK

RNA49,RNA59,RNA69,RNA48,NK TYPE

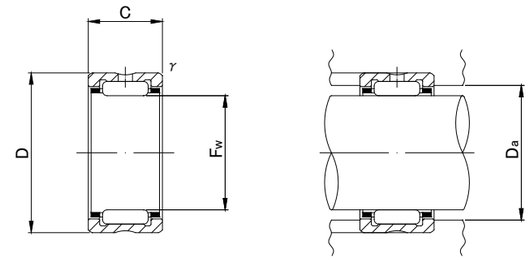
Shaft Diameter (mm)	Designation					Dimensions (mm)				Standard mounting dimensions (mm)	Basic dynamic load rating	Basic static load rating	Limiting speed *	Mass	Usable bearing designation	
	RNA 49	RNA 59	RNA 69	RNA 48	NK	Fw	D	C	r's min	Da MAX	Cr N	Cor N	rpm	g (approx)	INNER RING	WITH INNER RING
5	—	—	—	—	NK5/10	5 +0.018	10	10	0.15	6.5	2 420	1 950	40 000	3.4	—	—
	—	—	—	—	NK5/12	5 +0.010	10	12	0.15	6.5	3 080	2 660	40 000	4.2	—	—
6	—	—	—	—	NK6/10	6 +0.018	12	10	0.015	7.5	2 700	2 320	37 000	5.3	—	—
	—	—	—	—	NK6/12	6 +0.010	12	12	0.15	7.5	3 440	3 170	37 000	6.4	—	—
7	RNA 495	—	—	—	—	7	13	10	0.15	8.5	2 960	2 690	34 000	5.9	IR5710	NA495
	—	—	—	—	NK7/10	7 +0.022	14	10	0.3	8.5	3 600	2 960	34 000	6.9	—	—
	—	—	—	—	NK7/12	7 +0.013	14	12	0.3	8.5	4 610	4 050	34 000	8.3	—	—
8	RNA 496	—	—	—	—	8	15	10	0.15	13.8	3 900	3 400	32 000	7.3	IR6810	NA496
	—	—	—	—	NK8/12	8 +0.022	15	12	0.3	13	5 100	4 700	32 000	9	IR5812	NKI 5/12
	—	—	—	—	NK8/16	8 +0.013	15	16	0.3	13	7 100	7 300	32 000	13	IR5816	NKI 5/16
9	—	—	—	—	NK9/12	9	16	12	0.3	14	5 500	5 300	30 000	10	IR6912	NKI 6/12
	—	—	—	—	NK9/16	9 +0.022	16	16	0.3	14	7 600	8 200	30 000	13.2	IR6916	NKI 6/16
	RNA 497	—	—	—	—	9 +0.013	17	10	0.15	15.8	4 500	3 600	30 000	9.3	IR7910	NA497
10	—	—	—	—	NK10/12	10	17	12	0.3	15	5 900	6 000	28 000	10.7	IR71012	NKI 7/12
	—	—	—	—	NK10/16	10 +0.022	17	16	0.3	15	8 200	9 200	28 000	14.3	IR71016	NKI 7/16
	RNA 498	—	—	—	—	10 +0.013	19	11	0.2	17.4	6 200	5 000	28 000	12.6	IR81011	NA498
12	—	—	—	—	NK12/12	12	19	12	0.3	17	6 600	7 300	26 000	12.2	IR91212	NKI 9/12
	—	—	—	—	NK12/16	12 +0.027	19	16	0.3	17	9 200	11 200	26 000	16.3	IR91216	NKI 9/16
	RNA 499	—	—	—	—	12 +0.016	20	11	0.3	18	6 600	6 300	26 000	13.6	IR91211	NA499

* Suitable for oil lubrication. In case of grease lubrication, down to 60% of this value.

MACHINED RING NEEDLE ROLLER BEARINGS
WITHOUT INNER RING







RNA49,RNA59,RNA69(Fw ≦ 35),NK

RNA49,RNA59,RNA69,RNA48,NK TYPE

Shaft Diameter (mm)	Designation					Dimensions (mm)				Standard mounting dimensions (mm)	Basic dynamic load rating	Basic static load rating	Limiting speed *	Mass	Usable bearing designation	
	RNA 49	RNA 59	RNA 69	RNA 48	NK	Fw	D	C	r's min	Da MAX	Cr N	Cor N	rpm	g (approx)	INNER RING	WITH INNER RING
14	RNA 4900	—	—	—	—	14 +0.027 +0.016	22	13	0.3	20	9 200	10 100	24 000	16.5	IR101413	NA4900
	—	—	—	—	NK14/16	14 +0.027 +0.016	22	16	0.3	20	11 800	13 700	24 000	21	IR101416	NKI 10/16
	—	—	—	—	NK14/20	14 +0.027 +0.016	22	20	0.3	20	14 800	18 500	24 000	26.5	IR101420	NKI 10/20
15	—	—	—	—	NK15/16	15 +0.027 +0.016	23	16	0.3	21	12 400	14 900	23 000	22.5	—	—
	—	—	—	—	NK15/20	15 +0.027 +0.016	23	20	0.3	21	15 600	20 200	23 000	28	—	—
16	RNA 4901	—	—	—	—	16	24	13	0.3	22	9 700	11 100	23 000	18.1	IR121613	NA4901
	—	—	—	—	NK16/16	16 +0.027 +0.016	24	16	0.3	22	12 300	15 100	23 000	23	IR121616	NKI 12/16
	—	—	—	—	NK16/20	16 +0.027 +0.016	24	20	0.3	22	15 600	20 400	23 000	29	IR121620	NKI 12/20
	—	—	RNA 6901	—	—	16	24	22	0.3	22	17 100	23 000	23 000	30	IR121622	NA6901
17	—	—	—	—	NK17/16	17 +0.027 +0.016	25	16	0.3	23	12 800	16 300	22 000	24.5	—	—
	—	—	—	—	NK17/20	17 +0.027 +0.016	25	20	0.3	23	16 300	22 100	22 000	30.5	—	—
18	—	—	—	—	NK18/16	18 +0.027 +0.016	26	16	0.3	24	13 400	17 500	21 000	25.5	—	—
	—	—	—	—	NK18/20	18 +0.027 +0.016	26	20	0.3	24	17 000	23 600	21 000	32	—	—
19	—	—	—	—	NK19/16	19 +0.033 +0.020	27	16	0.3	25	14 000	18 700	21 000	27	IR151916	NKI 15/16
	—	—	—	—	NK19/20	19 +0.033 +0.020	27	20	0.3	25	17 700	25 300	21 000	34	IR151920	NKI 15/20
20	RNA 4902	—	—	—	—	20	28	13	0.3	26	10 900	13 800	20 000	21.5	IR152013	NA4902
	—	—	—	—	NK20/16	20	28	16	0.3	26	13 900	18 700	20 000	27.5	—	—
	—	RNA 5902	—	—	—	20 +0.033 +0.020	28	18	0.3	26	15 700	22 100	20 000	33	IR152018	NA5902
	—	—	—	—	NK20/20	20	28	20	0.3	26	17 600	25 400	20 000	35.5	—	—
	—	—	RNA 6902	—	—	20	28	23	0.3	26	19 300	28 700	20 000	37	IR152023	NA6902

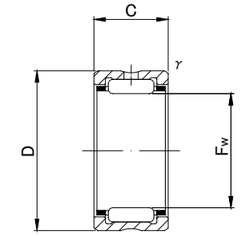
* Suitable for oil lubrication. In case of grease lubrication, down to 60% of this value.

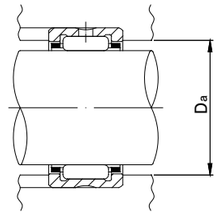
RNA, NK

MACHINED RING NEEDLE ROLLER BEARINGS
WITHOUT INNER RING









RNA49,RNA59,RNA69(Fw ≦ 35),NK

RNA49,RNA59,RNA69,RNA48,NK TYPE

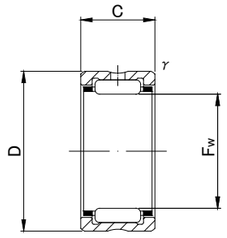
Shaft Diameter (mm)	Designation					Dimensions (mm)				Standard mounting dimensions (mm)	Basic dynamic load rating	Basic static load rating	Limiting speed *	Mass	Usable bearing designation	
	RNA 49	RNA 59	RNA 69	RNA 48	NK	Fw	D	C	r's min	Da MAX	Cr N	Cor N	rpm	g (approx)	INNER RING	WITH INNER RING
21	—	—	—	—	NK21/16	21 +0.033 +0.020	29	16	0.3	27	14 400	20 000	19 000	29	IR172116	NKI 17/16
	—	—	—	—	NK21/20	21 +0.033 +0.020	29	20	0.3	27	18 200	27 100	19 000	36	IR172120	NKI 17/20
22	RNA 4903	—	—	—	—	22 +0.033 +0.020	30	13	0.3	28	11 800	15 600	18 000	23.5	IR172213	NA4903
	—	—	—	—	NK22/16	22 +0.033 +0.020	30	16	0.3	28	14 900	21 200	18 000	30	—	—
	—	RNA 5903	—	—	—	22 +0.033 +0.020	30	18	0.3	28	16 900	24 900	18 000	35	IR172218	NA5903
	—	—	—	—	NK22/20	22 +0.033 +0.020	30	20	0.3	28	18 900	28 700	18 000	37.5	—	—
	—	—	RNA 6903	—	—	22 +0.033 +0.020	30	23	0.3	28	20 800	32 500	18 000	40.5	IR172223	NA6903
24	—	—	—	—	NK24/16	24 +0.033 +0.020	32	16	0.3	30	15 300	22 600	17 000	32	IR202416	NKI 20/16
	—	—	—	—	NK24/20	24 +0.033 +0.020	32	20	0.3	30	19 400	30 500	17 000	40.5	IR202420	NKI 20/20
25	—	—	—	—	NK25/16	25 +0.033 +0.020	33	16	0.3	31	15 800	23 700	16 000	33.5	—	—
	—	—	—	—	NK25/20	25 +0.033 +0.020	33	20	0.3	31	20 000	32 200	16 000	42	—	—
	RNA 4904	—	—	—	—	25 +0.033 +0.020	37	17	0.3	35	21 000	25 000	16 000	55.5	IR202517	NA4904
	—	RNA 5904	—	—	—	25 +0.033 +0.020	37	23	0.3	35	29 400	38 600	16 000	84	IR202523	NA5904
	—	—	RNA 6904	—	—	25 +0.033 +0.020	37	30	0.3	35	35 400	48 800	16 000	95.5	IR202530	NA6904
26	—	—	—	—	NK26/16	26 +0.033 +0.020	34	16	0.3	32	16 300	24 900	15 000	34.5	IR222616	NKI 22/16
	—	—	—	—	NK26/20	26 +0.033 +0.020	34	20	0.3	32	20 600	33 700	15 000	43.5	IR222620	NKI 22/20
28	—	—	—	—	NK28/20	28 +0.033 +0.020	37	20	0.3	35	21 700	37 100	14 000	51.5	—	—
	—	—	—	—	NK28/30	28 +0.033 +0.020	37	30	0.3	35	31 100	58 900	14 000	83.5	—	—
	RNA 49/22	—	—	—	—	28 +0.033 +0.020	39	17	0.3	37	21 400	28 800	14 000	56.5	IR222817	NA49/22
	—	RNA 59/22	—	—	—	28 +0.033 +0.020	39	23	0.3	37	29 800	44 400	14 000	92	IR222823	NA59/22
	—	—	RNA 69/22	—	—	28 +0.033 +0.020	39	30	0.3	37	36 300	56 900	14 000	97.5	IR222830	NA69/22
29	—	—	—	—	NK29/20	29 +0.033 +0.020	38	20	0.3	36	21 600	37 200	14 000	57	IR252920	NKI 25/20
	—	—	—	—	NK29/30	29 +0.033 +0.020	38	30	0.3	36	30 900	59 000	14 000	85	IR252930	NKI 25/30

* Suitable for oil lubrication. In case of grease lubrication, down to 60% of this value.

MACHINED RING NEEDLE ROLLER BEARINGS
WITHOUT INNER RING

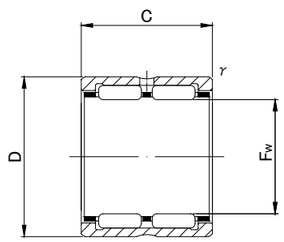


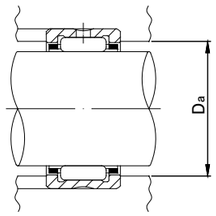




RNA49,RNA59,RNA69(Fw ≦ 35),NK







RNA69

RNA49,RNA59,RNA69,RNA48,NK TYPE

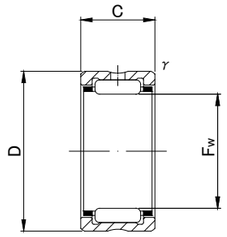
Shaft Diameter (mm)	Designation					Dimensions (mm)				Standard mounting dimensions (mm)	Basic dynamic load rating	Basic static load rating	Limiting speed *	Mass	Usable bearing designation	
	RNA 49	RNA 59	RNA 69	RNA 48	NK	Fw	D	C	r's min	Da MAX	Cr N	Cor N	rpm	g (approx)	INNER RING	WITH INNER RING
30	—	—	—	—	**NK30/20**	30	40	20	0.3	38	25 100	40 100	13 000	64.5	—	—
	—	—	—	—	**NK30/30**	30	40	30	0.3	38	36 000	63 800	13 000	97.5	—	—
	RNA 4905	—	—	—	—	30 +0.033 +0.020	42	17	0.3	40	23 700	30 700	13 000	64	IR253017	NA4905
	—	**RNA 5905**	—	—	—	30	42	23	0.3	40	33 200	47 500	13 000	101	IR253023	NA5905
	—	—	**RNA 6905**	—	—	30	42	30	0.3	40	42 100	64 200	13 000	111	IR253030	NA6905
32	—	—	—	—	**NK32/20**	32	42	20	0.3	40	25 700	42 200	13 000	68	IR283220	NKI 28/20
	—	—	—	—	**NK32/30**	32	42	30	0.3	40	36 900	67 100	13 000	102	IR283230	NKI 28/30
	RNA 49/28	—	—	—	—	32 +0.041 +0.025	45	17	0.3	43	24 500	32 700	13 000	76.5	IR283217	NA49/28
	—	**RNA 59/28**	—	—	—	32	45	23	0.3	43	34 300	50 500	13 000	108	IR283223	NA59/28
	—	—	**RNA 69/28**	—	—	32	45	30	0.3	43	41 800	64 700	13 000	133	IR283230	NA69/28
35	—	—	—	—	**NK35/20**	35	45	20	0.3	43	27 000	46 200	11 000	73.5	IR303520	NKI 30/20
	—	—	—	—	**NK35/30**	35	45	30	0.3	43	38 600	73 500	11 000	112	IR303530	NKI 30/30
	RNA 4906	—	—	—	—	35 +0.041 +0.025	47	17	0.3	45	25 200	34 700	11 000	72.5	IR303517	NA4906
	—	**RNA 5906**	—	—	—	35	47	23	0.3	45	35 200	53 700	11 000	108	IR303523	NA5906
	—	—	**RNA 6906**	—	—	35	47	30	0.3	45	43 100	69 000	11 000	125	IR303530	NA6906
37	—	—	—	—	**NK37/20**	37 +0.041 +0.025	47	20	0.3	45	28 200	50 100	11 000	77.5	IR323720	NKI 32/20
	—	—	—	—	**NK37/30**	37	47	30	0.3	45	40 500	79 800	11 000	117	IR323730	NKI 32/30
38	—	—	—	—	**NK38/20**	38 +0.041 +0.025	48	20	0.3	46	28 100	50 200	11 000	79	—	—
	—	—	—	—	**NK38/30**	38	48	30	0.3	46	40 300	80 000	11 000	119	—	—
40	—	—	—	—	**NK40/20**	40	50	20	0.3	48	29 400	54 100	10 000	83	IR354020	NKI 35/20
	—	—	—	—	**NK40/30**	40	50	30	0.3	48	42 300	86 100	10 000	125	IR354030	NKI 35/30
	RNA 49/32	—	—	—	—	40 +0.041 +0.025	52	20	0.6	48	31 300	47 900	10 000	96	IR324020	NA49/32
	—	**RNA 59/32**	—	—	—	40	52	27	0.6	48	41 900	69 900	10 000	149	IR324027	NA59/32
	—	—	**RNA 69/32**	—	—	40	52	36	0.6	48	53 500	95 700	10 000	172	IR324036	NA69/32
42	—	—	—	—	**NK42/20**	42	52	20	0.3	50	29 900	56 200	9 500	86.5	—	—
	—	—	—	—	**NK42/30**	42	52	30	0.3	50	43 000	89 500	9 500	130	—	—
	RNA 4907	—	—	—	—	42 +0.041 +0.025	55	20	0.6	51	32 000	50 200	9 500	113	IR354220	NA4907
	—	**RNA 5907**	—	—	—	42	55	27	0.6	51	42 900	73 200	9 500	176	IR354227	NA5907
	—	—	**RNA 6907**	—	—	42	55	36	0.6	51	54 800	100 000	9 500	200	IR354236	NA6907

* Suitable for oil lubrication. In case of grease lubrication, down to 60% of this value.

MACHINED RING NEEDLE ROLLER BEARINGS
WITHOUT INNER RING

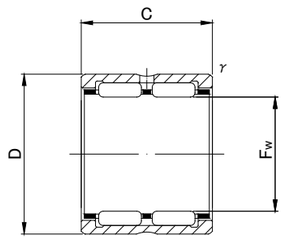


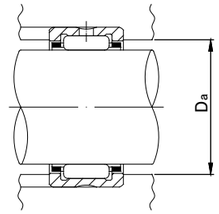




RNA49,RNA59,RNA48,NK







RNA69

RNA49,RNA59,RNA69,RNA48,NK TYPE

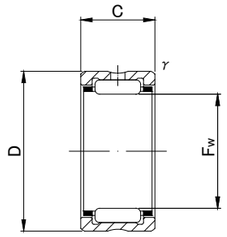
Shaft Diameter (mm)	Designation					Dimensions (mm)				Standard mounting dimensions (mm)	Basic dynamic load rating	Basic static load rating	Limiting speed *	Mass	Usable bearing designation	
	RNA 49	RNA 59	RNA 69	RNA 48	NK	Fw	D	C	r's min	Da MAX	Cr N	Cor N	rpm	g (approx)	INNER RING	WITH INNER RING
43	—	—	—	—	NK43/20	43 +0.041	53	20	0.3	51	30 500	58 100	9 500	88.5	IR384320	NKI 38/20
	—	—	—	—	NK43/30	43 +0.025	53	30	0.3	51	43 700	92 500	9 500	133	IR384330	NKI 38/30
45	—	—	—	—	NK45/20	45 +0.041	55	20	0.3	53	31 100	60 100	9 000	92	IR404520	NKI 40/20
	—	—	—	—	NK45/30	45 +0.025	55	30	0.3	53	44 500	95 700	9 000	138	IR404530	NKI 40/30
47	—	—	—	—	NK47/20	47 +0.041	57	20	0.3	55	31 500	62 300	8 500	95	IR424720	NKI 42/20
	—	—	—	—	NK47/30	47 +0.025	57	30	0.3	55	45 200	99 000	8 500	144	IR424730	NKI 42/30
48	RNA 4908	—	—	—	—	48	62	22	0.6	58	41 600	67 400	8 500	152	IR404822	NA4908
	—	RNA 5908	—	—	—	48 +0.041 +0.025	62	30	0.6	58	58 000	103 000	8 500	225	IR404830	NA5908
	—	—	RNA 6908	—	—	48	62	40	0.6	58	71 300	134 400	8 500	275	IR404840	NA6908
50	—	—	—	—	NK50/25	50 +0.041	62	25	0.6	58	43 000	85 200	8 000	159	IR455025	NKI 45/25
	—	—	—	—	NK50/35	50 +0.025	62	35	0.6	58	58 100	125 500	8 000	225	IR455035	NKI 45/35
52	RNA 4909	—	—	—	—	52	68	22	0.6	64	43 500	73 400	7 500	197	IR455222	NA4909
	—	RNA 5909	—	—	—	52 +0.049 +0.030	68	30	0.6	64	60 700	112 000	7 500	232	IR455230	NA5909
	—	—	RNA 6909	—	—	52	68	40	0.6	64	74 600	147 100	7 500	355	IR455240	NA6909
55	—	—	—	—	NK55/25	55 +0.049	68	25	0.6	64	45 400	94 100	7 500	193	IR505525	NKI 50/25
	—	—	—	—	NK55/35	55 +0.030	68	35	0.6	64	61 300	138 300	7 500	255	IR505535	NKI 50/35

* Suitable for oil lubrication. In case of grease lubrication, down to 60% of this value.

MACHINED RING NEEDLE ROLLER BEARINGS
WITHOUT INNER RING

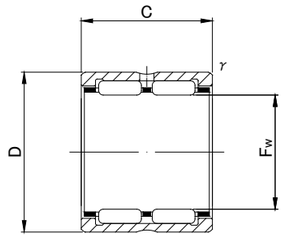


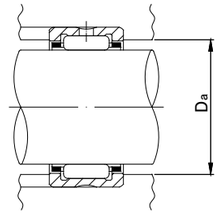




RNA49,RNA59,RNA48,NK







RNA69

RNA49,RNA59,RNA69,RNA48,NK TYPE

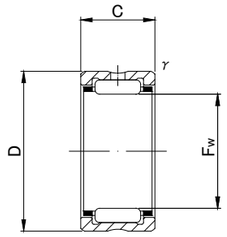
Shaft Diameter (mm)	Designation					Dimensions (mm)				Standard mounting dimensions (mm)	Basic dynamic load rating	Basic static load rating	Limiting speed *	Mass	Usable bearing designation	
	RNA 49	RNA 59	RNA 69	RNA 48	NK	Fw	D	C	r's min	Da MAX	Cr N	Cor N	rpm	g (approx)	INNER RING	WITH INNER RING
58	RNA 4910	—	—	—	—	58 +0.049 +0.030	72	22	0.6	68	46 200	82 100	7 000	179	IR505822	NA4910
	—	RNA 5910	—	—	—	58	72	30	0.6	68	64 400	126 000	7 000	289	IR505830	NA5910
	—	—	RNA 6910	—	—	58	72	40	0.6	68	79 100	163 800	7 000	320	IR505840	NA6910
60	—	—	—	—	NK60/25	60 +0.049 +0.030	72	25	0.6	68	47 500	103 000	6 500	187	IR556025	NKI 55/25
	—	—	—	—	NK60/35	60	72	35	0.6	68	64 100	151 000	6 500	260	IR556035	NKI 55/35
63	RNA 4911	—	—	—	—	63 +0.049 +0.030	80	25	1	75	57 600	97 300	6 500	265	IR556325	NA4911
	—	RNA 5911	—	—	—	63	80	34	1	75	82 600	154 000	6 500	367	IR556334	NA5911
	—	—	RNA 6911	—	—	63	80	45	1	75	99 000	194 200	6 500	475	IR556345	NA6911
65	—	—	—	—	NK65/25	65 +0.049 +0.030	78	25	0.6	74	49 600	111 800	6 000	225	—	—
	—	—	—	—	NK65/35	65	78	35	0.6	74	67 000	164 800	6 000	315	—	—
68	—	—	—	—	NK68/25	68 +0.049 +0.030	82	25	0.6	78	54 800	116 700	6 000	250	IR606825	NKI 60/25
	—	—	—	—	NK68/35	68	82	35	0.6	78	72 100	165 700	6 000	350	IR606835	NKI 60/35
	RNA 4912	—	—	—	—	68	85	25	1	80	60 100	104 900	6 000	285	IR606825	NA4912
	—	RNA 5912	—	—	—	68	85	34	1	80	86 100	167 000	6 000	408	IR606834	NA5912
	—	—	RNA 6912	—	—	68	85	45	1	80	103 000	210 800	6 000	510	IR606845	NA6912
70	—	—	—	—	NK70/25	70 +0.049 +0.030	85	25	0.6	81	55 500	120 600	5 500	280	—	—
	—	—	—	—	NK70/35	70	85	35	0.6	81	73 000	170 600	5 500	395	—	—
72	RNA 4913	—	—	—	—	72 +0.049 +0.030	90	25	1	85	62 800	113 800	5 500	325	IR657225	NA4913
	—	RNA 5913	—	—	—	72	90	34	1	85	89 900	180 000	5 500	462	IR657234	NA5913
	—	—	RNA 6913	—	—	72	90	45	1	85	107 900	226 500	5 500	585	IR657245	NA6913

* Suitable for oil lubrication. In case of grease lubrication, down to 60% of this value.

MACHINED RING NEEDLE ROLLER BEARINGS
WITHOUT INNER RING

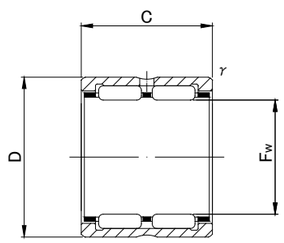


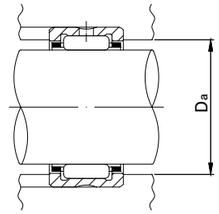




RNA49,RNA59,RNA48,NK







RNA69

RNA49,RNA59,RNA69,RNA48,NK TYPE

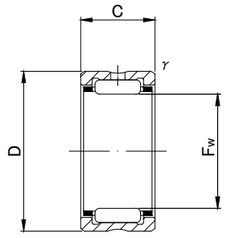
Shaft Diameter (mm)	Designation					Dimensions (mm)				Standard mounting dimensions (mm)	Basic dynamic load rating	Basic static load rating	Limiting speed *	Mass	Usable bearing designation	
	RNA 49	RNA 59	RNA 69	RNA 48	NK	Fw	D	C	r's min	Da MAX	Cr N	Cor N	rpm	g (approx)	INNER RING	WITH INNER RING
73	—	—	—	—	**NK73/25**	73 +0.049 +0.030	90	25	0.6	86	61 100	126 500	5 500	335	—	—
	—	—	—	—	**NK73/35**	73	90	35	0.6	86	80 400	180 400	5 500	475	IR657335	NKI 65/35
75	—	—	—	—	**NK75/25**	75 +0.049 +0.030	92	25	0.6	88	62 200	130 400	5 500	345	—	—
	—	—	—	—	**NK75/35**	75	92	35	0.6	88	82 700	186 300	5 500	485	—	—
80	—	—	—	—	**NK80/25**	80 +0.049 +0.030	95	25	1	90	59 400	137 300	5 000	315	IR708025	NKI 70/25
	—	—	—	—	**NK80/35**	80	95	35	1	90	78 100	194 200	5 000	445	IR708035	NKI 70/35
	RNA 4914	—	—	—	—	80	100	30	1	95	83 200	157 900	5 000	495	IR708030	NA4914
	—	RNA 5914	—	—	—	80	100	40	1	95	112 000	232 000	5 000	706	IR708040	NA5914
	—	—	RNA 6914	—	—	80	100	54	1	95	133 400	310 900	5 000	910	IR708054	NA6914
85	—	—	—	—	**NK85/25**	85 +0.058 +0.036	105	25	1	100	76 400	145 100	4 500	435	IR758525	NKI 75/25
	RNA 4915	—	—	—	—	85	105	30	1	100	86 200	169 700	4 500	525	IR758530	NA4915
	—	—	—	—	**NK85/35**	85	105	35	1	100	102 000	209 900	4 500	610	IR758535	NKI 75/35
	—	RNA 5915	—	—	—	85	105	40	1	100	116 000	249 000	4 500	745	IR758540	NA5915
	—	—	RNA 6915	—	—	85	105	54	1	100	138 300	330 500	4 500	960	IR758554	NA6915
90	—	—	—	—	**NK90/25**	90 +0.058 +0.036	110	25	1	105	77 400	150 000	4 500	456	IR809025	NKI 80/25
	RNA 4916	—	—	—	—	90	110	30	1	105	87 400	174 600	4 500	550	IR809030	NA4916
	—	—	—	—	**NK90/35**	90	110	35	1	105	103 000	216 700	4 500	640	IR809035	NKI 80/35
	—	RNA 5916	—	—	—	90	110	40	1	105	117 000	257 000	4 500	787	IR809040	NA5916
	—	—	RNA 6916	—	—	90	110	54	1	105	143 200	350 100	4 500	1 010	IR809054	NA6916
95	—	—	—	—	**NK95/26**	95 +0.058 +0.036	115	26	1	110	79 700	158 900	4 200	495	IR859526	NKI 85/26
	—	—	—	—	**NK95/36**	95	115	36	1	110	106 900	230 500	4 200	690	IR859536	NKI 85/36
100	—	—	—	—	**NK100/26**	100 +0.058 +0.036	120	26	1	115	82 500	168 700	4 000	525	IR9010026	NKI 90/26
	RNA 4917	—	—	—	—	100	120	35	1.1	113.5	109 800	244 200	4 000	705	IR8510035	NA4917
	—	—	—	—	**NK100/36**	100	120	36	1	115	109 800	244 200	4 000	725	IR9010036	NKI 90/36
	—	RNA 5917	—	—	—	100	120	46	1.1	113.5	144 000	346 000	4 000	1 000	IR8510046	NA5917
	—	—	RNA 6917	—	—	100	120	63	1.1	113.5	172 600	466 800	4 000	1 300	IR8510063	NA6917

* Suitable for oil lubrication. In case of grease lubrication, down to 60% of this value.

MACHINED RING NEEDLE ROLLER BEARINGS
WITHOUT INNER RING

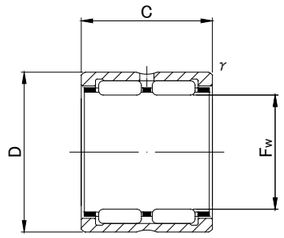


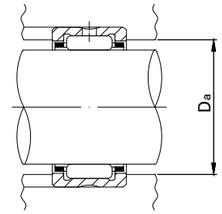




RNA49,RNA59,RNA48,NK







RNA69

RNA49,RNA59,RNA69,RNA48,NK TYPE

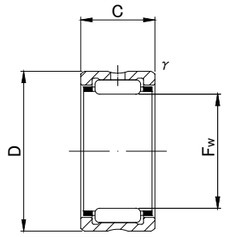
Shaft Diameter (mm)	Designation					Dimensions (mm)				Standard mounting dimensions (mm)	Basic dynamic load rating	Basic static load rating	Limiting speed *	Mass	Usable bearing designation	
	RNA 49	RNA 59	RNA 69	RNA 48	NK	Fw	D	C	r's min	Da MAX	Cr N	Cor N	rpm	g (approx)	INNER RING	WITH INNER RING
105	—	—	—	—	NK105/26	105	125	26	1	120	84 700	177 500	3 800	545	IR9510526	NKI 95/26
	RNA 4918	—	—	—	—	105	125	35	1.1	118.5	112 800	257 900	3 800	740	IR9010535	NA4918
	—	—	—	—	NK105/36	105 +0.058 +0.036	125	36	1	120	112 800	257 900	3 800	760	IR9510536	NKI 95/36
	—	RNA 5918	—	—	—	105	125	46	1.1	118.5	148 000	365 000	3 800	1 040	IR9010546	NA5918
	—	—	RNA 6918	—	—	105	125	63	1.1	118.5	177 500	490 300	3 800	1 360	IR9010563	NA6918
110	—	—	—	—	NK110/30	110	130	30	1.1	123.5	105 900	239 300	3 600	660	IR10011030	NKI 100/30
	RNA 4919	—	—	—	—	110	130	35	1.1	123.5	116 700	270 700	3 600	770	IR9511035	NA4919
	—	—	—	—	NK110/40	110 +0.058 +0.036	130	40	1.1	123.5	133 400	323 600	3 600	880	IR10011040	NKI 100/40
	—	RNA 5919	—	—	—	110	130	46	1.1	123.5	152 000	384 000	3 600	1 130	IR9511046	NA5919
	—	—	RNA 6919	—	—	110	130	63	1.1	1 123.5	182 400	514 800	3 600	1 420	IR9511063	NA6919
115	RNA 4920	—	—	—	—	115 +0.058 +0.036	140	40	1.1	133.5	145 000	329 000	3 500	1 190	IR10011540	NA4920
120	—	—	—	RNA 4822	—	120 +0.058 +0.036	140	30	1	135	93 000	239 000	3 500	790	IR11012030	NA4822
125	RNA 4922	—	—	—	—	125 +0.068 +0.043	150	40	1.1	143.5	152 000	357 000	3 000	1 280	IR11012540	NA4922
130	—	—	—	RNA 4824	—	130 +0.068 +0.043	150	30	1	145	97 000	259 000	3 000	850	IR12013030	NA4824
135	RNA 4924	—	—	—	—	135 +0.068 +0.043	165	45	1.1	158.5	187 000	435 000	3 000	1 930	IR12013545	NA4924

* Suitable for oil lubrication. In case of grease lubrication, down to 60% of this value.

MACHINED RING NEEDLE ROLLER BEARINGS
WITHOUT INNER RING

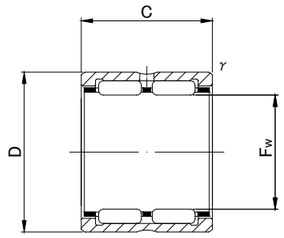


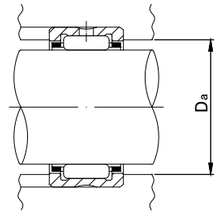




RNA49,RNA59,RNA48,NK







RNA69

RNA49,RNA59,RNA69,RNA48,NK TYPE

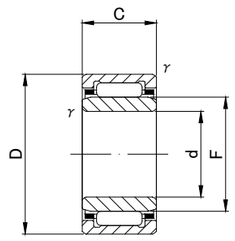
Shaft Diameter (mm)	Designation					Dimensions (mm)				Standard mounting dimensions (mm)	Basic dynamic load rating	Basic static load rating	Limiting speed *	Mass	Usable bearing designation	
	RNA 49	RNA 59	RNA 69	RNA 48	NK	Fw	D	C	r's min	Da MAX	Cr N	Cor N	rpm	g (approx)	INNER RING	WITH INNER RING
145	—	—	—	**RNA 4826**	—	145 +0.068 +0.043	165	35	1.1	158.5	117 000	340 000	3 000	1 100	IR13014535	NA4826
150	**RNA 4926**	—	—	—	—	150 +0.068 +0.043	180	50	1.5	172	216 000	540 000	2 500	2 360	IR13015050	NA4926
155	—	—	—	**RNA 4828**	—	155 +0.068 +0.043	175	35	1.1	168.5	121 000	363 000	2 500	1 170	IR14015535	NA4828
160	**RNA 4928**	—	—	—	—	160 +0.068 +0.043	190	50	1.5	182	224 000	580 000	2 500	2 500	IR14016050	NA4928
165	—	—	—	**RNA 4830**	—	165 +0.068 +0.043	190	40	1.1	183.5	168 000	446 000	2 500	1 750	IR15016540	NA4830
175	—	—	—	**RNA 4832**	—	175 +0.068 +0.043	200	40	1.1	193.5	173 000	474 000	2 500	1 850	IR16017540	NA4832

* Suitable for oil lubrication. In case of grease lubrication, down to 60% of this value.

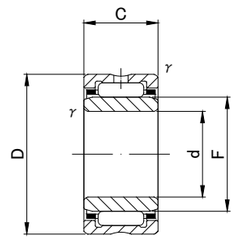
RNA, NK

MACHINED RING NEEDLE ROLLER BEARINGS
WITH INNER RING

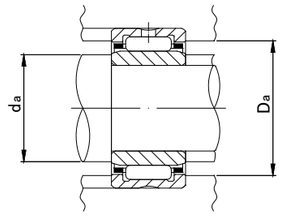




NKI(d ≦ 8)



NA49,NA59,NA69(d ≦ 30),NKI



NA49,NA59,NA69,NA48,NKI TYPE

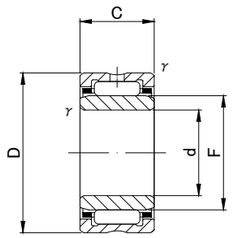
Shaft Diameter (mm)	Designation					Dimensions (mm)					Standard mounting dimensions (mm)			Basic dynamic load rating	Basic static load rating	Limiting speed*	Mass	Usable bearing designation	
	NA 49	NA 59	NA 69	NA48	NKI	d	D	C	r's min	F	da MIN	da MAX	Da MAX	Cr N	Cor N	rpm	g (approx)	OUTER RING	INNER RING
5	NA495	—	—	—	—	5 0 -0.008	13	10	0.15	7	6.2	6.7	11.8	2 960	2 690	34 000	7.3	RNA495	IR5710
	—	—	—	—	NKI 5/12	5 0 -0.008	15	12	0.3	8	7	7.7	13	5 100	4 700	32 000	11.9	NK8/12	IR5812
	—	—	—	—	NKI 5/16	5 0 -0.008	15	16	0.3	8	7	7.7	13	7 100	7 300	32 000	16.7	NK8/16	IR5816
6	NA 496	—	—	—	—	6 0 -0.008	15	10	0.15	8	7.2	7.7	13.8	3 900	3 400	32 000	9.1	RNA496	IR6810
	—	—	—	—	NKI 6/12	6 0 -0.008	16	12	0.3	9	8	8.7	14	5 500	5 300	30 000	13	NK9/12	IR6912
	—	—	—	—	NKI 6/16	6 0 -0.008	16	16	0.3	9	8	8.7	14	7 600	8 200	30 000	17.5	NK9/16	IR6916
7	NA 497	—	—	—	—	7 0 -0.008	17	10	0.15	9	8.2	8.7	15.8	4 500	3 600	30 000	11.2	RNA497	IR7910
	—	—	—	—	NKI 7/12	7 0 -0.008	17	12	0.3	10	9	9.7	15	5 900	6 000	28 000	14.3	NK10/12	IR71012
	—	—	—	—	NKI 7/16	7 0 -0.008	17	16	0.3	10	9	9.7	15	8 200	9 200	28 000	19.2	NK10/16	IR71016
8	NA 498	—	—	—	—	8 0 -0.008	19	11	0.2	10	9.2	9.7	17.4	6 200	5 000	28 000	15	RNA498	IR81011
9	—	—	—	—	NKI 9/12	9 0 -0.008	19	12	0.3	12	11	11.5	17	6 600	7 300	26 000	16.7	NK12/12	IR91212
	—	—	—	—	NKI 9/16	9 0 -0.008	19	16	0.3	12	11	11.5	17	9 200	11 200	26 000	22.5	NK12/16	IR91216
	NA 499	—	—	—	—	9 0 -0.008	20	11	0.3	12	11	11.5	18	6 600	6 300	26 000	16.7	RNA499	IR91211
10	NA 4900	—	—	—	—	10 0 -0.008	22	13	0.3	14	12	13	20	9 200	10 100	24 000	24	RNA4900	IR101413
	—	—	—	—	NKI 10/16	10 0 -0.008	22	16	0.3	14	12	13	20	11 800	13 700	24 000	30	NK14/16	IR101416
	—	—	—	—	NKI 10/20	10 0 -0.008	22	20	0.3	14	12	13	20	14 800	18 500	24 000	38	NK14/20	IR101420
12	NA 4901	—	—	—	—	12 0 -0.008	24	13	0.3	16	14	15	22	9 700	11 100	23 000	26.5	RNA4901	IR121613
	—	—	—	—	NKI 12/16	12 0 -0.008	24	16	0.3	16	14	15	22	12 300	15 100	23 000	33.5	NK16/16	IR121616
	—	—	—	—	NKI 12/20	12 0 -0.008	24	20	0.3	16	14	15	22	15 600	20 400	23 000	42.5	NK16/20	IR121620
	—	—	NA 6901	—	—	12 0 -0.008	24	22	0.3	16	14	15	22	17 100	23 000	23 000	44.5	RNA6901	IR121622

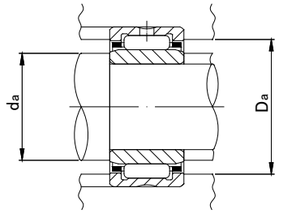
* Suitable for oil lubrication. In case of grease lubrication, down to 60% of this value.

MACHINED RING NEEDLE ROLLER BEARINGS
WITH INNER RING









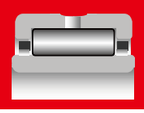
NA49,NA59,NA69(d ≦ 30),NKI

NA49,NA59,NA69,NA48,NKI TYPE

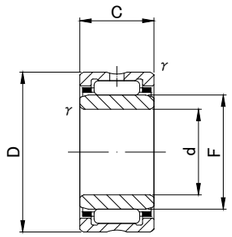
Shaft Diameter (mm)	Designation					Dimensions (mm)					Standard mounting dimensions (mm)			Basic dynamic load rating	Basic static load rating	Limiting speed*	Mass	Usable bearing designation	
	NA 49	NA 59	NA 69	NA48	NKI	d	D	C	r's min	F	da MIN	da MAX	Da MAX	Cr N	Cor N	rpm	g (approx)	OUTER RING	INNER RING
15	—	—	—	—	NKI 15/16	15	27	16	0.3	19	17	18	25	14 000	18 700	21 000	39.5	NK19/16	IR151916
	—	—	—	—	NKI 15/20	15	27	20	0.3	19	17	18	25	17 700	25 300	21 000	50	NK19/20	IR151920
	NA 4902	—	—	—	—	15 0 -0.008	28	13	0.3	20	17	19	26	10 900	13 800	20 000	35	RNA4902	IR152013
	—	NA 5902	—	—	—	15	28	18	0.3	20	17	19	26	15 700	22 100	20 000	52	RNA5902	IR152018
	—	—	NA 6902	—	—	15	28	23	0.3	20	17	19	26	19 300	28 700	20 000	61	RNA6902	IR152023
17	—	—	—	—	NKI 17/16	17	29	16	0.3	21	19	20	27	14 400	20 000	19 000	43.5	NK21/16	IR172116
	—	—	—	—	NKI 17/20	17	29	20	0.3	21	19	20	27	18 200	27 100	19 000	54	NK21/20	IR172120
	NA 4903	—	—	—	—	17 0 -0.008	30	13	0.3	22	19	21	28	11 800	15 600	18 000	39	RNA4903	IR172213
	—	NA 5903	—	—	—	17	30	18	0.3	22	19	21	28	16 900	24 900	18 000	56	RNA5903	IR172218
	—	—	NA 6903	—	—	17	30	23	0.3	22	19	21	28	20 800	32 500	18 000	67	RNA6903	IR172223
20	—	—	—	—	NKI 20/16	20	32	16	0.3	24	22	23	30	15 300	22 600	17 000	48.5	NK24/16	IR202416
	—	—	—	—	NKI 20/20	20	32	20	0.3	24	22	23	30	19 400	30 500	17 000	61	NK24/20	IR202420
	NA 4904	—	—	—	—	20 0 -0.010	37	17	0.3	25	22	24	35	21 000	25 000	16 000	78.5	RNA4904	IR202517
	—	NA 5904	—	—	—	20	37	23	0.3	25	22	24	35	29 400	38 600	16 000	115	RNA5904	IR202523
	—	—	NA 6904	—	—	20	37	30	0.3	25	22	24	35	35 400	48 800	16 000	136	RNA6904	IR202530
22	—	—	—	—	NKI 22/16	22	34	16	0.3	26	24	25	32	16 300	24 900	15 000	52	NK26/16	IR222616
	—	—	—	—	NKI 22/20	22	34	20	0.3	26	24	25	32	20 600	33 700	15 000	67.5	NK26/20	IR222620
	NA 49/22	—	—	—	—	22 0 -0.010	39	17	0.3	28	24	27	37	21 400	28 800	14 000	87	RNA49/22	IR222817
	—	NA 59/22	—	—	—	22	39	23	0.3	28	24	27	37	29 800	44 400	14 000	134	RNA59/22	IR222823
	—	—	NA 69/22	—	—	22	39	30	0.3	28	24	27	37	36 300	56 900	14 000	152	RNA69/22	IR222830
25	—	—	—	—	NKI 25/20	25	38	20	0.3	29	27	28	36	21 600	37 200	14 000	82	NK29/20	IR252920
	—	—	—	—	NKI 25/30	25	38	30	0.3	29	27	28	36	30 900	59 000	14 000	123	NK29/30	IR252930
	NA 4905	—	—	—	—	25 0 -0.010	42	17	0.3	30	27	29	40	23 700	30 700	13 000	92.5	RNA4905	IR253017
	—	NA 5905	—	—	—	25	42	23	0.3	30	27	29	40	33 200	47 500	13 000	139	RNA5905	IR253023
	—	—	NA 6905	—	—	25	42	30	0.3	30	27	29	40	42 100	64 200	13 000	160	RNA6905	IR253030
28	—	—	—	—	NKI 28/20	28	42	20	0.3	32	30	31	40	25 700	42 200	13 000	96.5	NK32/20	IR283220
	—	—	—	—	NKI 28/30	28	42	30	0.3	32	30	31	40	36 900	67 100	13 000	145	NK32/30	IR283230
	NA 49/28	—	—	—	—	28 0 -0.010	45	17	0.3	32	30	31	43	24 500	32 700	13 000	101	RNA49/28	IR283217
	—	NA 59/28	—	—	—	28	45	23	0.3	32	30	31	43	34 300	50 500	13 000	142	RNA59/28	IR283223
	—	—	NA 69/28	—	—	28	45	30	0.3	32	30	31	43	41 800	64 700	13 000	176	RNA69/28	IR283230
30	—	—	—	—	NKI 30/20	30	45	20	0.3	35	32	34	43	27 000	46 200	11 000	112	NK35/20	IR303520
	—	—	—	—	NKI 30/30	30	45	30	0.3	35	32	34	43	38 600	73 500	11 000	171	NK35/30	IR303530
	NA 4906	—	—	—	—	30 0 -0.010	47	17	0.3	35	32	34	45	25 200	34 700	11 000	106	RNA4906	IR303517
	—	NA 5906	—	—	—	30	47	23	0.3	35	32	34	45	35 200	53 700	11 000	152	RNA5906	IR303523
	—	—	NA 6906	—	—	30	47	30	0.3	35	32	34	45	43 100	69 000	11 000	184	RNA6906	IR303530

* Suitable for oil lubrication. In case of grease lubrication, down to 60% of this value.

MACHINED RING NEEDLE ROLLER BEARINGS
WITH INNER RING

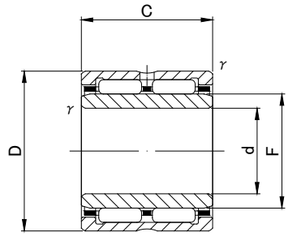




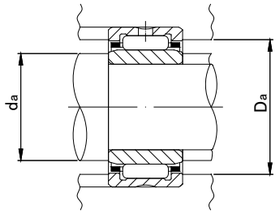


NA49,NA59,NA69(d ≦ 30),NKI





NA69



NA, NKI

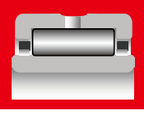
NA49,NA59,NA69,NA48,NKI TYPE

Shaft Diameter (mm)	Designation					Dimensions (mm)					Standard mounting dimensions (mm)			Basic dynamic load rating	Basic static load rating	Limiting speed*	Mass	Usable bearing designation	
	NA 49	NA 59	NA 69	NA48	NKI	d	D	C	r's min	F	da MIN	da MAX	Da MAX	Cr N	Cor N	rpm	g (approx)	OUTER RING	INNER RING
32	—	—	—	—	NKI 32/20	32	47	20	0.3	37	34	36	45	28 200	50 100	11 000	121	NK37/20	IR323720
	—	—	—	—	NKI 32/30	32	47	30	0.3	37	34	36	45	40 500	79 800	11 000	180	NK37/30	IR323730
	NA 49/32	—	—	—	—	32	52	20	0.6	40	36	39	48	31 300	47 900	10 000	165	RNA49/32	IR324020
	—	NA 59/32	—	—	—	32	52	27	0.6	40	36	39	48	41 900	69 900	10 000	241	RNA59/32	IR324027
	—	—	NA 69/32	—	—	32	52	36	0.6	40	36	39	48	53 500	95 700	10 000	295	RNA69/32	IR324036
35	—	—	—	—	NKI 35/20	35	50	20	0.3	40	37	39	48	29 400	54 100	10 000	129	NK40/20	IR354020
	—	—	—	—	NKI 35/30	35	50	30	0.3	40	37	39	48	42 300	86 100	10 000	192	NK40/30	IR354030
	NA 4907	—	—	—	—	35	55	20	0.6	42	39	41	51	32 000	50 200	9 500	178	RNA4907	IR354220
	—	NA 5907	—	—	—	35	55	27	0.6	42	39	41	51	42 900	73 200	9 500	256	RNA5907	IR354227
	—	—	NA 6907	—	—	35	55	36	0.6	42	39	41	51	54 800	100 000	9 500	320	RNA6907	IR354236
38	—	—	—	—	NKI 38/20	38	53	20	0.3	43	40	42	51	30 500	58 100	9 500	136	NK43/20	IR384320
	—	—	—	—	NKI 38/30	38	53	30	0.3	43	40	42	51	43 700	92 500	9 500	205	NK43/30	IR384330
40	—	—	—	—	NKI 40/20	40	55	20	0.3	45	42	44	53	31 100	60 100	9 000	143	NK45/20	IR404520
	—	—	—	—	NKI 40/30	40	55	30	0.3	45	42	44	53	44 500	95 700	9 000	215	NK45/30	IR404530
	NA 4908	—	—	—	—	40	62	22	0.6	48	44	47	58	41 600	67 400	8 500	245	RNA4908	IR404822
	—	NA 5908	—	—	—	40	62	30	0.6	48	44	47	58	58 000	103 000	8 500	348	RNA5908	IR404830
	—	—	NA 6908	—	—	40	62	40	0.6	48	44	47	58	71 300	134 400	8 500	440	RNA6908	IR404840
42	—	—	—	—	NKI 42/20	42	57	20	0.3	47	44	46	55	31 500	62 300	8 500	149	NK47/20	IR424720
	—	—	—	—	NKI 42/30	42	57	30	0.3	47	44	46	55	45 200	99 000	8 500	225	NK47/30	IR424730
45	—	—	—	—	NKI 45/25	45	62	25	0.6	50	49	49.5	58	43 000	85 200	8 000	230	NK50/25	IR455025
	—	—	—	—	NKI 45/35	45	62	35	0.6	50	49	49.5	58	58 100	125 500	8 000	320	NK50/35	IR455035
	NA 4909	—	—	—	—	45	68	22	0.6	52	49	51	64	43 500	73 400	7 500	285	RNA4909	IR455222
	—	NA 5909	—	—	—	45	68	30	0.6	52	49	51	64	60 700	112 000	7 500	396	RNA5909	IR455230
	—	—	NA 6909	—	—	45	68	40	0.6	52	49	51	64	74 600	147 100	7 500	520	RNA6909	IR455240
50	—	—	—	—	NKI 50/25	50	68	25	0.6	55	54	54.5	64	45 400	94 100	7 500	270	NK55/25	IR505525
	—	—	—	—	NKI 50/35	50	68	35	0.6	55	54	54.5	64	61 300	138 300	7 500	365	NK55/35	IR505535
	NA 4910	—	—	—	—	50	72	22	0.6	58	54	57	68	46 200	82 100	7 000	295	RNA4910	IR505822
	—	NA 5910	—	—	—	50	72	30	0.6	58	54	57	68	64 400	126 000	7 000	498	RNA5910	IR505830
	—	—	NA 6910	—	—	50	72	40	0.6	58	54	57	68	79 100	163 800	7 000	530	RNA6910	IR505840

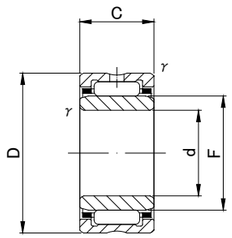
Dimensions (mm) d: 0 -0.012

* Suitable for oil lubrication. In case of grease lubrication, down to 60% of this value.

MACHINED RING NEEDLE ROLLER BEARINGS
WITH INNER RING

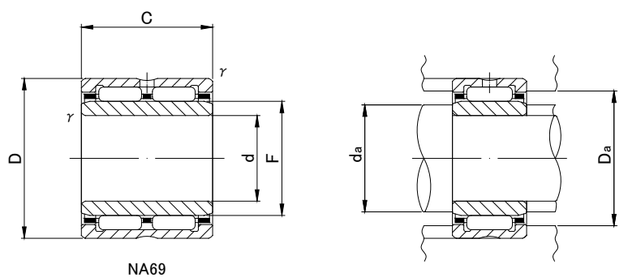






NA49,NA59,NKI





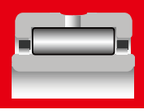
NA69

NA49,NA59,NA69,NA48,NKI TYPE

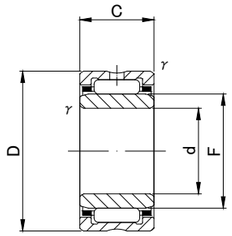
Shaft Diameter (mm)	Designation					Dimensions (mm)					Standard mounting dimensions (mm)			Basic dynamic load rating	Basic static load rating	Limiting speed*	Mass	Usable bearing designation	
	NA 49	NA 59	NA 69	NA48	NKI	d	D	C	r's min	F	da MIN	da MAX	Da MAX	Cr N	Cor N	rpm	g (approx)	OUTER RING	INNER RING
55	—	—	—	—	NKI 55/25	55 0 -0.015	72	25	0.6	60	59	59.5	68	47 500	103 000	6 500	275	NK60/25	IR556025
	—	—	—	—	NKI 55/35	55	72	35	0.6	60	59	59.5	68	64 100	151 000	6 500	380	NK60/35	IR556035
	NA 4911	—	—	—	—	55	80	25	1	63	60	61	75	57 600	97 300	6 500	410	RNA4911	IR556325
	—	NA 5911	—	—	—	55	80	34	1	63	60	61	75	82 600	154 000	6 500	559	RNA5911	IR556334
	—	—	NA 6911	—	—	55	80	45	1	63	60	61	75	99 000	194 200	6 500	730	RNA6911	IR556345
60	—	—	—	—	NKI 60/25	60 0 -0.015	82	25	0.6	68	64	66	78	54 800	116 700	6 000	395	NK68/25	IR606825
	—	—	—	—	NKI 60/35	60	82	35	0.6	68	64	66	78	72 100	165 700	6 000	560	NK68/35	IR606835
	NA 4912	—	—	—	—	60	85	25	1	68	64	66	80	60 100	104 900	6 000	440	RNA4912	IR606825
	—	NA 5912	—	—	—	60	85	34	1	68	65	66	80	86 100	167 000	6 000	614	RNA5912	IR606834
	—	—	NA 6912	—	—	60	85	45	1	68	65	66	80	103 000	210 800	6 000	785	RNA6912	IR606845
65	NA 4913	—	—	—	—	65 0 -0.015	90	25	1	72	70	70.5	85	62 800	113 800	5 500	470	RNA4913	IR657225
	—	NA 5913	—	—	—	65	90	34	1	72	70	70.5	85	89 900	180 000	5 500	655	RNA5913	IR657234
	—	—	—	—	NKI 65/35	65	90	35	0.6	73	69	71	86	80 400	180 400	5 500	710	NK73/35	IR657335
	—	—	NA 6913	—	—	65	90	45	1	72	70	70.5	85	107 900	226 500	5 500	840	RNA6913	IR657245
70	—	—	—	—	NKI 70/25	70 0 -0.015	95	25	1	80	75	78	90	59 400	137 300	5 000	540	NK80/25	IR708025
	—	—	—	—	NKI 70/35	70	95	35	1	80	75	78	90	78 100	194 200	5 000	755	NK80/35	IR708035
	NA 4914	—	—	—	—	70	100	30	1	80	75	78	95	83 200	157 900	5 000	765	RNA4914	IR708030
	—	NA 5914	—	—	—	70	100	40	1	80	75	78	95	112 000	232 000	5 000	1 060	RNA5914	IR708040
	—	—	NA 6914	—	—	70	100	54	1	80	75	78	95	133 400	310 900	5 000	1 400	RNA6914	IR708054
75	—	—	—	—	NKI 75/25	75 0 -0.015	105	25	1	85	80	83	100	76 400	145 100	4 500	675	NK85/25	IR758525
	NA 4915	—	—	—	—	75	105	30	1	85	80	83	100	86 200	169 700	4 500	810	RNA4915	IR758530
	—	—	—	—	NKI 75/35	75	105	35	1	85	80	83	100	102 000	209 900	4 500	945	NK85/35	IR758535
	—	NA 5915	—	—	—	75	105	40	1	85	80	83	100	116 000	249 000	4 500	1 130	RNA5915	IR758540
	—	—	NA 6915	—	—	75	105	54	1	85	80	83	100	138 300	330 500	4 500	1 480	RNA6915	IR758554
80	—	—	—	—	NKI 80/25	80 0 -0.015	110	25	1	90	85	88	105	77 400	150 000	4 500	710	NK90/25	IR809025
	NA 4916	—	—	—	—	80	110	30	1	90	85	88	105	87 400	174 600	4 500	855	RNA4916	IR809030
	—	—	—	—	NKI 80/35	80	110	35	1	90	85	88	105	103 000	216 700	4 500	995	NK90/35	IR809035
	—	NA 5916	—	—	—	80	110	40	1	90	85	88	105	117 000	257 000	4 500	1 150	RNA5916	IR809040
	—	—	NA 6916	—	—	80	110	54	1	90	85	88	105	143 200	350 100	4 500	1 560	RNA6916	IR809054
85	—	—	—	—	NKI 85/26	85 0 -0.020	115	26	1	95	90	93	110	79 700	158 900	4 200	775	NK95/26	IR859526
	—	—	—	—	NKI 85/36	85	115	36	1	95	90	93	110	106 900	230 500	4 200	1 080	NK95/36	IR859536
	NA 4917	—	—	—	—	85	120	35	1.1	100	91.5	98	133.5	109 800	244 200	4 000	1 280	RNA4917	IR8510035
	—	NA 5917	—	—	—	85	120	46	1.1	100	91.5	98	133.5	144 000	346 000	4 000	1 760	RNA5917	IR8510046
	—	—	NA 6917	—	—	85	120	63	1.1	100	91.5	98	133.5	172 600	466 800	4 000	2 340	RNA6917	IR8510063

* Suitable for oil lubrication. In case of grease lubrication, down to 60% of this value.

MACHINED RING NEEDLE ROLLER BEARINGS
WITH INNER RING

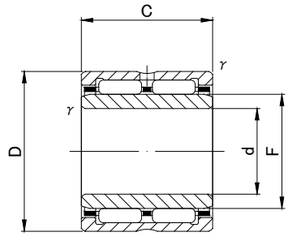


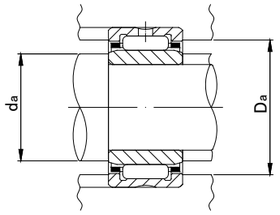




NA49,NA59,NA48,NKI







NA69

NA49,NA59,NA69,NA48,NKI TYPE

Shaft Diameter (mm)	Designation					Dimensions (mm)					Standard mounting dimensions (mm)			Basic dynamic load rating	Basic static load rating	Limiting speed*	Mass	Usable bearing designation	
	NA 49	NA 59	NA 69	NA48	NKI	d	D	C	*r*s min	F	da MIN	da MAX	Da MAX	Cr N	Cor N	rpm	g (approx)	OUTER RING	INNER RING
90	—	—	—	—	**NKI 90/26**	90 0 -0.020	120	26	1	100	95	98	135	82 500	168 700	4 000	820	NK100/26	IR9010026
	—	—	—	—	**NKI 90/36**		120	36	1	100	95	98	135	109 800	244 200	4 000	1 140	NK100/36	IR9010036
	NA 4918	—	—	—	—		125	35	1.1	105	96.5	103	118.5	112 800	257 900	3 800	1 350	RNA4918	IR9010535
	—	**NA 5918**	—	—	—		125	46	1.1	105	96.5	103	118.5	148 000	365 000	3 800	1 840	RNA5918	IR9010546
	—	—	**NA 6918**	—	—		125	63	1.1	105	96.5	103	118.5	177 500	490 300	3 800	2 460	RNA6918	IR9010563
95	—	—	—	—	**NKI 95/26**	95 0 -0.020	125	26	1	105	100	103	120	84 700	177 500	3 800	860	NK105/26	IR9510526
	—	—	—	—	**NKI 95/36**		125	36	1	105	100	103	120	112 800	257 900	3 800	1 190	NK105/36	IR9510536
	NA 4919	—	—	—	—		130	35	1.1	110	101.5	108	123.5	116 700	270 700	3 600	1 420	RNA4919	IR9511035
	—	**NA 5919**	—	—	—		130	46	1.1	110	101.5	108	123.5	152 000	384 000	3 600	1 980	RNA5919	IR9511046
	—	—	**NA 6919**	—	—		130	63	1.1	110	101.5	108	123.5	182 400	514 800	3 600	2 580	RNA6919	IR9511063
100	—	—	—	—	**NKI 100/30**	100 0 -0.020	130	30	1.1	110	106.5	108	123.5	105 900	239 300	3 600	1 040	NK110/30	IR10011030
	—	—	—	—	**NKI 100/40**		130	40	1.1	110	106.5	108	123.5	133 400	323 600	3 600	1 380	NK110/40	IR10011040
	NA 4920	—	—	—	—		140	40	1.1	115	106.5	113	133.5	145 000	329 000	3 500	1 960	RNA4920	IR10011540
110	—	—	—	**NA 4822**	—	110 0 -0.020	140	30	1	120	115	118	135	93 000	239 000	3 500	1 200	RNA4822	IR11012030
	NA 4922	—	—	—	—	110	150	40	1.1	125	116.5	123	143.5	152 000	357 000	3 000	2 120	RNA4922	IR11012540
120	—	—	—	**NA 4824**	—	120 0 -0.020	150	30	1	130	125	128	145	97 000	259 000	3 000	1 300	RNA4824	IR12013030
	NA 4924	—	—	—	—	120	165	45	1.1	135	126.5	133	158.5	187 000	435 000	3 000	2 960	RNA4924	IR12013545
130	—	—	—	**NA 4826**	—	130 0 -0.025	165	35	1.1	145	136.5	143	158.5	117 000	340 000	3 000	1 960	RNA4826	IR13014535
	NA 4926	—	—	—	—	130	180	50	1.5	150	138	148	172	216 000	540 000	2 500	4 030	RNA4926	IR13015050
140	—	—	—	**NA 4828**	—	140 0 -0.025	175	35	1.1	155	146.5	153	168.5	121 000	363 000	2 500	2 100	RNA4828	IR14015535
	NA 4928	—	—	—	—	140	190	50	1.5	160	148	158	182	224 000	580 000	2 500	4 290	RNA4928	IR14016050
150	—	—	—	**NA 4830**	—	150 0 -0.025	190	40	1.1	165	156.5	163	183.5	168 000	446 000	2 500	2 880	RNA4830	IR15016540
160	—	—	—	**NA 4832**	—	160 0 -0.025	200	40	1.1	175	166.5	173	193.5	173 000	474 000	2 500	3 050	RNA4832	IR16017540

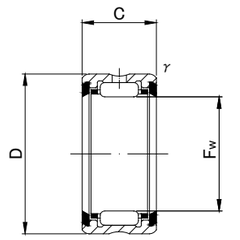
* Suitable for oil lubrication. In case of grease lubrication, down to 60% of this value.

MACHINED RING NEEDLE ROLLER BEARINGS

SEALED, WITHOUT INNER RING

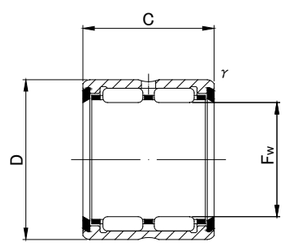


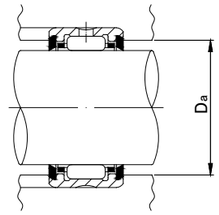




RNA49UU,RNA69UU（Fw ≦ 35）







RNA69UU

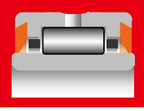
RNA49UU,RNA69UU,SEALED TYPE

Prepacked Grease

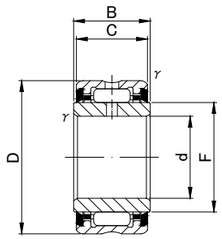
Shaft Diameter (mm)	Designation		Dimensions (mm)				Standard mounting dimensions (mm)	Basic dynamic load rating	Basic static load rating	Limiting speed (GREASE)	Mass	Usable bearing designation	
	RNA 49UU	RNA 69UU	Fw	D	C	r's min	Da MAX	Cr N	Cor N	rpm	g (approx)	INNER RING	WITH INNER RING
14	RNA 4900UU	—	14 +0.027 +0.016	22	13	0.3	20	8 000	8 500	14 000	16	IRZ101414	NA4900UU
16	RNA 4901UU	—	16 +0.027	24	13	0.3	22	8 400	9 300	12 000	18	IRZ121614	NA4901UU
	—	RNA 6901UU	16 +0.016	24	22	0.3	22	15 600	20 400	12 000	30	IRZ121623	NA6901UU
20	RNA 4902UU	—	20 +0.033	28	13	0.3	26	9 600	11 700	9 500	22	IRZ152014	NA4902UU
	—	RNA 6902UU	20 +0.020	28	23	0.3	26	18 400	27 100	9 500	38	IRZ152024	NA6902UU
22	RNA 4903UU	—	22 +0.033	30	13	0.3	28	10 300	13 100	8 500	23	IRZ172214	NA4903UU
	—	RNA 6903UU	22 +0.020	30	23	0.3	28	19 800	30 600	8 500	40	IRZ172224	NA6903UU
25	RNA 4904UU	—	25 +0.033	37	17	0.3	35	17 900	20 500	7 500	55	IRZ202518	NA4904UU
	—	RNA 6904UU	25 +0.020	37	30	0.3	35	33 000	44 500	7 500	96	IRZ202531	NA6904UU
30	RNA 4905UU	—	30 +0.033	42	17	0.3	40	20 300	25 100	6 500	63	IRZ253018	NA4905UU
	—	RNA 6905UU	30 +0.020	42	30	0.3	40	39 200	58 600	6 500	110	IRZ253031	NA6905UU
35	RNA 4906UU	—	35 +0.041	47	17	0.3	45	21 600	28 400	5 500	71	IRZ303518	NA4906UU
	—	RNA 6906UU	35 +0.025	47	30	0.3	45	40 100	63 100	5 500	130	IRZ303531	NA6906UU
42	RNA 4907UU	—	42 +0.041	55	20	0.6	51	30 100	46 300	4 500	110	IRZ354221	NA4907UU
	—	RNA 6907UU	42 +0.025	55	36	0.6	51	51 600	92 600	4 500	200	IRZ354237	NA6907UU
48	RNA 4908UU	—	48 +0.041	62	22	0.6	58	37 200	58 300	4 000	150	IRZ404823	NA4908UU
	—	RNA 6908UU	48 +0.025	62	40	0.6	58	63 700	116 700	4 000	270	IRZ404841	NA6908UU
52	RNA 4909UU	—	52 +0.049	68	22	0.6	64	38 800	63 400	3 500	190	IRZ455223	NA4909UU
	—	RNA 6909UU	52 +0.030	68	40	0.6	64	66 700	126 500	3 500	355	IRZ455241	NA6909UU
58	RNA 4910UU	—	58 +0.049	72	22	0.6	68	41 300	71 100	3 500	180	IRZ505823	NA4910UU
	—	RNA 6910UU	58 +0.030	72	40	0.6	68	70 800	142 200	3 500	320	IRZ505841	NA6910UU

MACHINED RING NEEDLE ROLLER BEARINGS

SEALED, WITH INNER RING

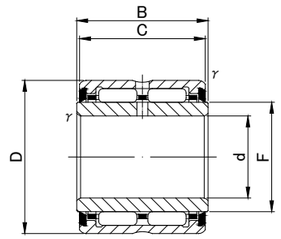




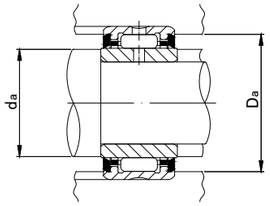


NA49UU,NA69UU(d ≦ 30)





NA69UU

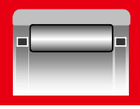


NA49UU,NA69UU,SEALED TYPE

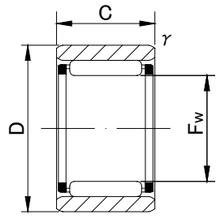
Prepacked Grease

Shaft Diameter (mm)	Designation		Dimensions (mm)						Standard mounting dimensions (mm)			Basic dynamic load rating	Basic static load rating	Limiting speed (GREASE)	Mass	Usable bearing designation	
	NA 49UU	NA 69UU	d	D	C	B	*r*s min	F	da MIN	da MAX	Da MAX	Cr N	Cor N	rpm	g (approx)	OUTER RING	INNER RING
10	NA 4900UU	—	10 0 -0.008	22	13	14	0.3	14	12	13	20	8 000	8 500	14 000	25	RNA4900UU	IRZ101414
12	NA 4901UU	—	12 0 -0.008	24	13	14	0.3	16	14	15	22	8 400	9 300	12 000	28	RNA4901UU	IRZ121614
	—	NA 6901UU	12	24	22	23	0.3	16	14	15	22	15 600	20 400	12 000	46	RNA6901UU	IRZ121623
15	NA 4902UU	—	15 0 -0.008	28	13	14	0.3	20	17	19	26	9 600	11 700	9 500	36	RNA4902UU	IRZ152014
	—	NA 6902UU	15	28	23	24	0.3	20	17	19	26	18 400	27 100	9 500	63	RNA6902UU	IRZ152024
17	NA 4903UU	—	17 0 -0.008	30	13	14	0.3	22	19	21	28	10 300	13 100	8 500	40	RNA4903UU	IRZ172214
	—	NA 6903UU	17	30	23	24	0.3	22	19	21	28	19 800	30 600	8 500	69	RNA6903UU	IRZ172224
20	NA 4904UU	—	20 0 -0.010	37	17	18	0.3	25	22	24	35	17 900	20 500	7 500	78	RNA4904UU	IRZ202518
	—	NA 6904UU	20	37	30	31	0.3	25	22	24	35	33 000	44 500	7 500	140	RNA6904UU	IRZ202531
25	NA 4905UU	—	25 0 -0.010	42	17	18	0.3	30	27	29	40	20 300	25 100	6 500	93	RNA4905UU	IRZ253018
	—	NA 6905UU	25	42	30	31	0.3	30	27	29	40	39 200	58 600	6 500	162	RNA6905UU	IRZ253031
30	NA 4906UU	—	30 0 -0.010	47	17	18	0.3	35	32	34	45	21 600	28 400	5 500	106	RNA4906UU	IRZ303518
	—	NA 6906UU	30	47	30	31	0.3	35	32	34	45	40 100	63 100	5 500	185	RNA6906UU	IRZ303531
35	NA 4907UU	—	35 0 -0.012	55	20	21	0.6	42	39	41	51	30 100	46 300	4 500	179	RNA4907UU	IRZ354221
	—	NA 6907UU	35	55	36	37	0.6	42	39	41	51	51 600	92 600	4 500	320	RNA6907UU	IRZ354237
40	NA 4908UU	—	40 0 -0.012	62	22	23	0.6	48	44	47	58	37 200	58 300	4 000	245	RNA4908UU	IRZ404823
	—	NA 6908UU	40	62	40	41	0.6	48	44	47	58	63 700	116 700	4 000	440	RNA6908UU	IRZ404841
45	NA 4909UU	—	45 0 -0.012	68	22	23	0.6	52	49	51	64	38 800	63 400	3 500	290	RNA4909UU	IRZ455223
	—	NA 6909UU	45	68	40	41	0.6	52	49	51	64	66 700	126 500	3 500	510	RNA6909UU	IRZ455241
50	NA 4910UU	—	50 0 -0.012	72	22	23	0.6	58	54	57	68	41 300	71 100	3 500	300	RNA4910UU	IRZ505823
	—	NA 6910UU	50	72	40	41	0.6	58	54	57	68	70 800	142 200	3 500	530	RNA6910UU	IRZ505841

MACHINED RING NEEDLE ROLLER BEARINGS
WITHOUT INNER RING

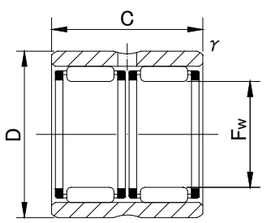




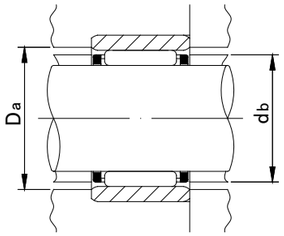


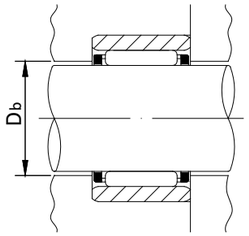
RNAF





RNAFW





RNAF(W) TYPE

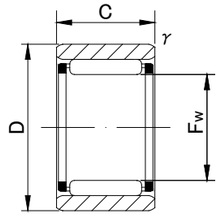
Shaft Diameter (mm)	Designation	Dimensions (mm)				Standard mounting dimensions (mm)			Basic dynamic load rating	Basic static load rating	Limiting speed*	Mass	Usable bearing designation	
	RNAF(W)	Fw	D	C	r s min	db	Da MAX	Db	Cr N	Cor N	rpm	g (approx)	INNER RING	WITH INNER RING
5	RNAF5108	5 +0.018 +0.010	10	8	0.2	6.7	8.4	5.4	2 500	2 000	40 000	3	—	—
6	RNAF6138	6 +0.018 +0.010	13	8	0.3	8.4	11	6.4	2 500	2 100	37 000	5.5	—	—
7	RNAF7148	7 +0.022 +0.013	14	8	0.3	9.4	12	7.4	2 900	2 600	34 000	6	—	—
8	RNAF81510	8 +0.022 +0.013	15	10	0.3	10.4	13	8.4	3 600	3 600	32 000	8	—	—
	RNAFW81620	8	16	20	0.3	10.8	14	8.4	6 200	7 200	32 000	20	—	—
10	RNAF101710	10 +0.022 +0.013	17	10	0.3	12.4	15	10.4	4 100	4 500	28 000	10	IR61010	NAF61710
	RNAF102012	10	20	12	0.3	13.5	18	10.4	6 000	6 000	28 000	19	IR71012	NAF72012
12	RNAF122212	12 +0.027 +0.016	22	12	0.3	17.5	20	12.4	9 000	8 400	26 000	19	IR91212	NAF92212
14	RNAF142213	14 +0.027 +0.016	22	13	0.3	17.6	20	14.6	7 800	9 400	24 000	18	IR101413	NAF102213
	RNAFW142220	14	22	20	0.3	17.6	20	14.6	10 800	14 200	24 000	28	IR101420	NAFW102220
	RNAF142612	14	26	12	0.3	19.4	24	14.6	9 800	9 700	24 000	29	IR101412	NAF102612
15	RNAF152313	15 +0.027 +0.016	23	13	0.3	18.6	21	15.6	8 200	10 200	23 000	20	—	—
	RNAFW152320	15	23	20	0.3	18.6	21	15.6	11 400	15 400	23 000	31	—	—
16	RNAF162413	16 +0.027 +0.016	24	13	0.3	19.6	22	16.6	8 600	11 000	23 000	21	IR121613	NAF122413
	RNAFW162420	16	24	20	0.3	19.6	22	16.6	11 900	16 700	23 000	32	IR121620	NAFW122420
	RNAF162812	16	28	12	0.3	21.4	26	16.6	10 500	10 900	23 000	32	IR121612	NAF122812
17	RNAF172513	17 +0.027 +0.016	25	13	0.3	20.6	23	17.6	9 000	11 900	22 000	22	—	—
	RNAFW172520	17	25	20	0.3	20.6	23	17.6	12 400	17 900	22 000	33	—	—
18	RNAF182613	18 +0.027 +0.016	26	13	0.3	21.6	24	18.6	9 300	12 700	21 000	23	—	—
	RNAFW182620	18	26	20	0.3	21.6	24	18.6	12 800	19 100	21 000	35	—	—
	RNAF183012	18	30	12	0.3	23.4	28	18.6	11 800	13 000	21 000	35	—	—
	RNAFW183024	18	30	24	0.3	23.4	28	18.6	20 200	26 200	21 000	70	—	—

* Suitable for oil lubrication. In case of grease lubrication, down to 50% of this value.

MACHINED RING NEEDLE ROLLER BEARINGS
WITHOUT INNER RING

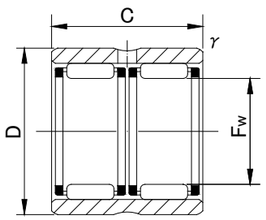




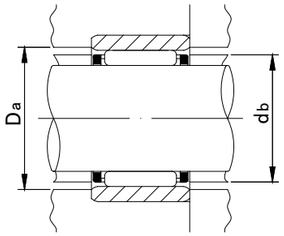


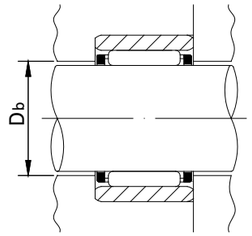
RNAF





RNAFW





RNAF(W) TYPE

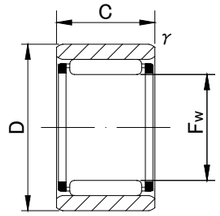
Shaft Diameter (mm)	Designation	Dimensions (mm)				Standard mounting dimensions (mm)			Basic dynamic load rating	Basic static load rating	Limiting speed*	Mass	Usable bearing designation	
	RNAF(W)	Fw	D	C	r s min	db	Da MAX	Db	Cr N	Cor N	rpm	g (approx)	INNER RING	WITH INNER RING
20	RNAF202813	20	28	13	0.3	23.6	26	20.6	9 600	13 500	20 000	25	IR152013	NAF152813
	RNAFW202826	20 +0.033 +0.020	28	26	0.3	23.6	26	20.6	16 500	27 100	20 000	50	IR152026	NAFW152826
	RNAF203212	20	32	12	0.3	25.4	30	20.6	12 400	14 300	20 000	37	IR152012	NAF153212
	RNAFW203224	20	32	24	0.3	25.4	30	20.6	21 200	28 600	20 000	75	—	—
22	RNAF223013	22	30	13	0.3	25.6	28	22.6	10 200	15 200	18 000	27	IR172213	NAF173013
	RNAFW223026	22 +0.033 +0.020	30	26	0.3	25.6	28	22.6	17 600	30 300	18 000	54	IR172226	NAFW173026
	RNAF223516	22	35	16	0.3	27.8	33	22.6	17 600	20 900	18 000	59	IR172216	NAF173516
	RNAFW223532	22	35	32	0.3	27.8	33	22.6	30 200	41 900	18 000	117	IR172232	NAFW173532
25	RNAF253517	25	35	17	0.3	29.5	33	25.6	17 300	26 600	16 000	51	IR202517	NAF203517
	RNAFW253526	25 +0.033 +0.020	35	26	0.3	29.5	33	25.6	22 500	37 200	16 000	78	IR202526	NAFW203526
	RNAF253716	25	37	16	0.3	30.4	35	25.6	19 400	24 500	16 000	57	IR202516	NAF203716
	RNAFW253732	25	37	32	0.3	30.4	35	25.6	33 200	49 000	16 000	114	IR202532	NAFW203732
28	RNAF284016	28 +0.033 +0.020	40	16	0.3	33.4	38	28.6	20 100	26 500	14 000	63	—	—
	RNAFW284032	28	40	32	0.3	33.4	38	28.6	34 400	53 000	14 000	125	—	—
30	RNAF304017	30	40	17	0.3	34.5	38	30.6	18 600	31 100	13 000	59	IR253017	NAF254017
	RNAFW304026	30 +0.033 +0.020	40	26	0.3	34.5	38	30.6	24 200	43 400	13 000	91	IR253026	NAFW254026
	RNAF304216	30	42	16	0.3	35.4	40	30.6	20 800	28 300	13 000	66	IR253016	NAF254216
	RNAFW304232	30	42	32	0.3	35.4	40	30.6	35 700	56 800	13 000	132	IR253032	NAFW254232
35	RNAF354517	35	45	17	0.3	39.5	43	35.6	20 500	36 900	11 000	68	IR303517	NAF304517
	RNAFW354526	35 +0.041 +0.025	45	26	0.3	39.5	43	35.6	26 600	51 600	11 000	103	IR303526	NAFW304526
	RNAF354716	35	47	16	0.3	40.4	45	35.6	23 000	33 800	11 000	76	IR303516	NAF304716
	RNAFW354732	35	47	32	0.3	40.4	45	35.6	39 500	67 800	11 000	151	IR303532	NAFW304732
40	RNAF405017	40	50	17	0.3	43.5	48	40.8	22 200	42 700	10 000	75	IR354017	NAF355017
	RNAFW405034	40 +0.041 +0.025	50	34	0.3	43.5	48	40.8	38 000	85 300	10 000	152	IR354034	NAFW355034
	RNAF405520	40	55	20	0.3	45.2	53	40.8	31 500	48 000	10 000	142	IR354020	NAF355520
	RNAFW405540	40	55	40	0.3	45.2	53	40.8	53 900	96 000	10 000	280	IR354040	NAFW355540
45	RNAF455517	45	55	17	0.3	48.5	53	45.8	23 200	47 200	10 000	84	IR404517	NAF405517
	RNAFW455534	45 +0.041 +0.025	55	34	0.3	48.5	53	45.8	39 900	94 200	10 000	167	IR404534	NAFW405534
	RNAF456220	45	62	20	0.3	50.9	60	45.8	33 200	53 300	9 000	185	IR404520	NAF406220
	RNAFW456240	45	62	40	0.3	50.9	60	45.8	57 000	106 900	9 000	370	IR404540	NAFW406240
50	RNAF506220	50	62	20	0.3	54.2	60	50.8	27 100	59 300	8 000	139	IR455020	NAF456220
	RNAFW506240	50 +0.041 +0.025	62	40	0.3	54.2	60	50.8	46 400	118 700	8 000	276	IR455040	NAFW456240
	RNAF506520	50	65	20	0.3	55.2	63	50.8	35 900	61 100	8 000	170	—	—
	RNAFW506540	50	65	40	0.6	55.2	61	50.8	61 500	122 600	8 000	345	—	—

* Suitable for oil lubrication. In case of grease lubrication, down to 50% of this value.

MACHINED RING NEEDLE ROLLER BEARINGS
WITHOUT INNER RING

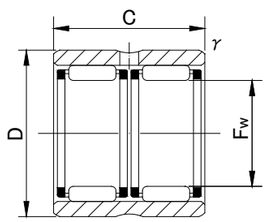




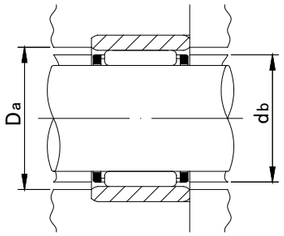


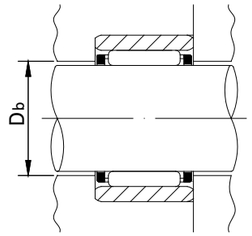
RNAF





RNAFW



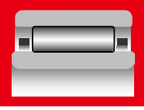


RNAF(W) TYPE

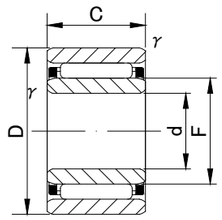
Shaft Diameter (mm)	Designation	Dimensions (mm)				Standard mounting dimensions (mm)			Basic dynamic load rating	Basic static load rating	Limiting speed*	Mass	Usable bearing designation	
	RNAF(W)	Fw	D	C	r s min	db	Da MAX	Db	Cr N	Cor N	rpm	g (approx)	INNER RING	WITH INNER RING
55	RNAF556820	55	68	20	0.6	59.5	64	55.8	28 500	66 000	7 500	167	IR505520	NAF506820
	RNAFW556840	55 +0.049 +0.030	68	40	0.6	59.5	64	55.8	48 900	132 400	7 500	330	IR505540	NAFW506840
	RNAF557220	55	72	20	0.6	60.9	68	55.8	37 400	66 400	7 500	215	IR455520	NAF457220
	RNAFW557240	55	72	40	0.6	60.9	68	55.8	64 100	132 400	7 500	435	IR455540	NAFW457240
60	RNAF607820	60 +0.049	78	20	1	66.3	73	60.8	38 900	71 700	6 500	255	IR506020	NAF507820
	RNAFW607840	60 +0.030	78	40	1	66.3	73	60.8	66 700	143 200	6 500	510	IR506040	NAFW507840
65	RNAF658530	65 +0.049	85	30	1	72	80	66	59 300	127 500	6 000	465	IR556530	NAF558530
	RNAFW658560	65 +0.030	85	60	1	72	80	66	102 000	254 000	6 000	950	IR556560	NAFW558560
70	RNAF709030	70 +0.049	90	30	1	77	85	71	61 200	135 300	5 500	500	IR607030	NAF609030
	RNAFW709060	70 +0.030	90	60	1	77	85	71	104 900	271 600	5 500	1 000	IR607060	NAFW609060
75	RNAF759530	75 +0.049	95	30	1	82	90	76	63 200	144 200	5 500	530	IR657530	NAF659530
	RNAFW759560	75 +0.030	95	60	1	82	90	76	108 900	289 300	5 500	1 050	IR657560	NAFW659560
80	RNAF8010030	80 +0.049	100	30	1	87	95	81	64 900	153 000	5 000	560	IR708030	NAF7010030
	RNAFW8010060	80 +0.030	100	60	1	87	95	81	111 800	306 000	5 000	1 120	IR708060	NAFW7010060
85	RNAF8510530	85 +0.058 +0.036	105	30	1	92	100	86	66 600	160 800	4 800	590	IR758530	NAF7510530
90	RNAF9011030	90 +0.058 +0.036	110	30	1	97	105	91	69 500	173 600	4 500	620	IR809030	NAF8011030
95	RNAF9511530	95 +0.058 +0.036	115	30	1	102	110	96	70 900	182 400	4 200	650	IR859530	NAF8511530
100	RNAF10012030	100 +0.058 +0.036	120	30	1	107	115	101	72 600	191 200	4 000	690	IR9010030	NAF9012030

* Suitable for oil lubrication. In case of grease lubrication, down to 50% of this value.

MACHINED RING NEEDLE ROLLER BEARINGS
WITH INNER RING

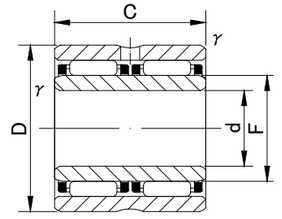


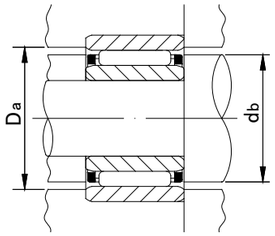


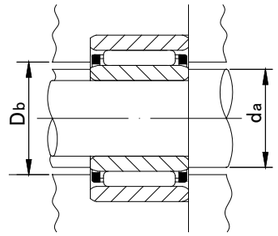


NAF









NAFW

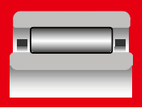
NAF(W)

NAF(W) TYPE

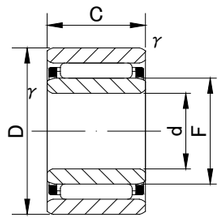
Shaft Diameter (mm)	Designation	Dimensions (mm)					Standard mounting dimensions (mm)					Basic dynamic load rating	Basic static load rating	Limiting speed*	Mass	Usable bearing designation	
	NAF(W)	d	D	C	r s min	F	db	Da MAX	da MIN	da MAX	Db	Cr N	Cor N	rpm	g (approx)	OUTER RING	INNER RING
6	NAF 61710	6 0 -0.008	17	10	0.3	10	12.4	15	8	9.7	10.4	4 100	4 500	28 000	14	RNAF101710	IR61010
7	NAF 72012	7 0 -0.008	20	12	0.3	10	13.5	18	9	9.7	10.4	6 000	6 000	28 000	23	RNAF102012	IR71012
9	NAF 92212	9 0 -0.008	22	12	0.3	12	17.5	20	11	11.5	12.4	9 000	8 400	26 000	24	RNAF122212	IR91212
10	NAF 102213	10 0 -0.008	22	13	0.3	14	17.6	20	12	13	14.6	7 800	9 400	24 000	26	RNAF142213	IR101413
	NAFW 102220	10	22	20	0.3	14	17.6	20	12	13	14.6	10 800	14 200	24 000	40	RNAFW142220	IR101420
	NAF 102612	10	26	12	0.3	14	19.4	24	12	13	14.6	9 800	9 700	24 000	36	RNAF142612	IR101412
12	NAF 122413	12 0 -0.008	24	13	0.3	16	19.6	22	14	15	16.6	8 600	11 000	23 000	30	RNAF162413	IR121613
	NAFW 122420	12	24	20	0.3	16	19.6	22	14	15	16.6	11 900	16 700	23 000	45	RNAFW162420	IR121620
	NAF 122812	12	28	12	0.3	16	21.4	26	14	15	16.6	10 500	10 900	23 000	40	RNAF162812	IR121612
15	NAF 152813	15 0 -0.008	28	13	0.3	20	23.6	26	17	19	20.6	9 600	13 500	20 000	37	RNAF202813	IR152013
	NAFW 152826	15	28	26	0.3	20	23.6	26	17	19	20.6	16 500	27 100	20 000	76	RNAFW202826	IR152026
	NAF 153212	15	32	12	0.3	20	25.4	30	17	19	20.6	12 400	14 300	20 000	51	RNAF203212	IR152012
17	NAF 173013	17 0 -0.008	30	13	0.3	22	25.6	28	19	21	22.6	10 200	15 200	18 000	43	RNAF223013	IR172213
	NAFW 173026	17	30	26	0.3	22	25.6	28	19	21	22.6	17 600	30 300	18 000	85	RNAFW223026	IR172226
	NAF 173516	17	35	16	0.3	22	27.8	33	19	21	22.6	17 600	20 900	18 000	77	RNAF223516	IR172216
	NAFW 173532	17	35	32	0.3	22	27.8	33	19	21	22.6	30 200	41 900	18 000	155	RNAFW223532	IR172232
20	NAF 203517	20 0 -0.010	35	17	0.3	25	29.5	33	22	24	25.6	17 300	26 600	16 000	75	RNAF253517	IR202517
	NAFW 203526	20	35	26	0.3	25	29.5	33	22	24	25.6	22 500	37 200	16 000	114	RNAFW253526	IR202526
	NAF 203716	20	37	16	0.3	25	30.4	35	22	24	25.6	19 400	24 500	16 000	80	RNAF253716	IR202516
	NAFW 203732	20	37	32	0.3	25	30.4	35	22	24	25.6	33 200	49 000	16 000	158	RNAFW253732	IR202532
25	NAF 254017	25 0 -0.010	40	17	0.3	30	34.5	38	27	29	30.6	18 600	31 100	13 000	88	RNAF304017	IR253017
	NAFW 254026	25	40	26	0.3	30	34.5	38	27	29	30.6	24 200	43 400	13 000	136	RNAFW304026	IR253026
	NAF 254216	25	42	16	0.3	30	35.4	40	27	29	30.6	20 800	28 300	13 000	94	RNAF304216	IR253016
	NAFW 254232	25	42	32	0.3	30	35.4	40	27	29	30.6	35 700	56 800	13 000	187	RNAFW304232	IR253032
30	NAF 304517	30 0 -0.010	45	17	0.3	35	39.5	43	32	34	35.6	20 500	36 900	11 000	100	RNAF354517	IR303517
	NAFW 304526	30	45	26	0.3	35	39.5	43	32	34	35.6	26 600	51 600	11 000	155	RNAFW354526	IR303526
	NAF 304716	30	47	16	0.3	35	40.4	45	32	34	35.6	23 000	33 800	11 000	108	RNAF354716	IR303516
	NAFW 304732	30	47	32	0.3	35	40.4	45	32	34	35.6	39 500	67 800	11 000	215	RNAFW354732	IR303532

* Suitable for oil lubrication. In case of grease lubrication, down to 50% of this value.

MACHINED RING NEEDLE ROLLER BEARINGS
WITH INNER RING

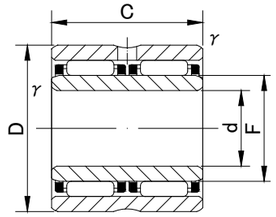




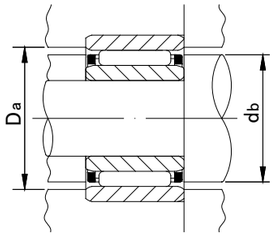


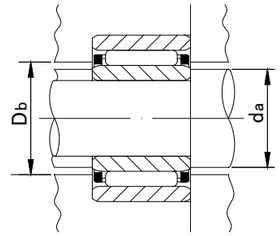
NAF





NAFW





NAF(W)

NAF(W) TYPE

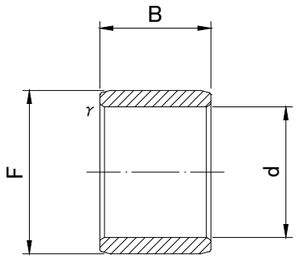
Shaft Diameter (mm)	Designation	Dimensions (mm)					Standard mounting dimensions (mm)					Basic dynamic load rating	Basic static load rating	Limiting speed*	Mass	Usable bearing designation	
	NAF(W)	d	D	C	r s min	F	db	Da MAX	da MIN	da MAX	Db	Cr N	Cor N	rpm	g (approx)	OUTER RING	INNER RING
35	NAF 355017	35	50	17	0.3	40	43.5	48	37	39	40.8	22 200	42 700	10 000	115	RNAF405017	IR354017
	NAFW 355034	35 0 -0.012	50	34	0.3	40	43.5	48	37	39	40.8	38 000	85 300	10 000	230	RNAFW405034	IR354034
	NAF 355520	35	55	20	0.3	40	45.2	53	37	39	40.8	31 500	48 000	10 000	188	RNAF405520	IR354020
	NAFW 355540	35	55	40	0.3	40	45.2	53	37	39	40.8	53 900	96 000	10 000	375	RNAFW405540	IR354040
40	NAF 405517	40	55	17	0.3	45	48.5	53	42	44	45.8	23 200	47 200	10 000	129	RNAF455517	IR404517
	NAFW 405534	40 0 -0.012	55	34	0.3	45	48.5	53	42	44	45.8	39 900	94 200	10 000	255	RNAFW455534	IR404534
	NAF 406220	40	62	20	0.3	45	50.9	60	42	44	45.8	33 200	53 300	9 000	236	RNAF456220	IR404520
	NAFW 406240	40	62	40	0.3	45	50.9	60	42	44	45.8	57 000	106 900	9 000	475	RNAFW456240	IR404540
45	NAF 456220	45	62	20	0.3	50	54.2	60	47	49	50.8	27 100	59 300	8 000	197	RNAF506220	IR455020
	NAFW 456240	45 0 -0.012	62	40	0.3	50	54.2	60	49	49.5	50.8	46 400	118 700	8 000	389	RNAFW506240	IR455040
	NAF 457220	45	72	20	0.6	55	60.9	68	49	54	55.8	37 400	66 400	7 500	340	RNAF557220	IR455520
	NAFW 457240	45	72	40	0.6	55	60.9	68	49	54	55.8	64 100	132 400	7 500	685	RNAFW557240	IR455540
50	NAF 506820	50	68	20	0.6	55	59.5	64	54	54.5	55.8	28 500	66 000	7 500	230	RNAF556820	IR505520
	NAFW 506840	50 0 -0.012	68	40	0.6	55	59.5	64	54	54.5	55.8	48 900	132 400	7 500	465	RNAFW556840	IR505540
	NAF 507820	50	78	20	1	60	66.3	73	55	59	60.8	38 900	71 700	6 500	390	RNAF607820	IR506020
	NAFW 507840	50	78	40	1	60	66.3	73	55	59	60.8	66 700	143 200	6 500	775	RNAFW607840	IR506040
55	NAF 558530	55 0 -0.015	85	30	1	65	72	80	60	63	66	59 300	127 500	6 000	680	RNAF658530	IR556530
	NAFW 558560	55	85	60	1	65	72	80	60	63	66	102 000	254 000	6 000	1 380	RNAFW658560	IR556560
60	NAF 609030	60 0 -0.015	90	30	1	70	77	85	65	68	71	61 200	135 300	5 500	740	RNAF709030	IR607030
	NAFW 609060	60	90	60	1	70	77	85	65	68	71	104 900	271 600	5 500	1 470	RNAFW709060	IR607060
65	NAF 659530	65 0 -0.015	95	30	1	75	82	90	70	73	76	63 200	144 200	5 500	800	RNAF759530	IR657530
	NAFW 659560	65	95	60	1	75	82	90	70	73	76	108 900	289 300	5 500	1 570	RNAFW759560	IR657560
70	NAF 7010030	70 0 -0.015	100	30	1	80	87	95	75	78	81	64 900	153 000	5 000	840	RNAF801003	IR708030
	NAFW 7010060	70	100	60	1	80	87	95	75	78	81	111 800	306 000	5 000	1 670	RNAFW8010060	IR708060
75	NAF 7510530	75 0 -0.015	105	30	1	85	92	100	80	83	86	66 600	160 800	4 800	890	RNAF8510530	IR758530
80	NAF 8011030	80 0 -0.015	110	30	1	90	97	105	85	88	91	69 500	173 600	4 500	930	RNAF9011030	IR809030
85	NAF 8511530	85 0 -0.020	115	30	1	95	102	110	90	93	96	70 900	182 400	4 200	970	RNAF9511530	IR859530
90	NAF 9012030	90 0 -0.020	120	30	1	100	107	115	95	98	101	72 600	191 200	4 000	1 040	RNAF10012030	IR9010030

* Suitable for oil lubrication. In case of grease lubrication, down to 50% of this value.

INNER RINGS

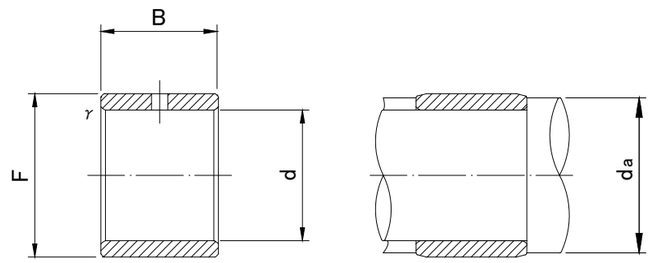






IR





IRZ

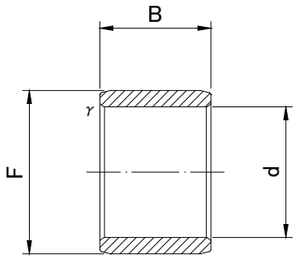
IR,IRZ TYPE

Shaft Diameter (mm)	Designation		Dimensions (mm)				Standard mounting dimensions (mm)		Mass	Usable bearing designation					
	IR	IRZ	d	F	B	*r*s min	da MIN	da MAX	g (approx)	RNA 49	RNA 59	RNA 69	RNA 48	NK	RNAF
5	**IR 5812**	—	5 0	8	12	0.3	7	7.7	3	—	—	—	—	NK 8/12	—
	IR 5816	—	5 -0.008	8	16	0.3	7	7.7	4	—	—	—	—	NK 8/16	—
6	**IR 6810**	—	6	8	10	0.15	7.2	7.7	1.7	RNA 496	—	—	—	—	—
	IR 6912	—	6 0	9	12	0.3	8	8.7	3.2	—	—	—	—	NK 9/12	—
	IR 6916	—	6 -0.008	9	16	0.3	8	8.7	4.3	—	—	—	—	NK 9/16	—
	IR 61010	—	6	10	10	0.3	8	9.7	4	—	—	—	—	—	RNAF 101710
7	**IR 7910**	—	7	9	10	0.15	8.2	8.7	1.9	RNA 497	—	—	—	—	—
	IR 71010.5	—	7	10	10.5	0.3	9	9.7	3.2	—	—	—	—	—	—
	IR 71012	—	7 0	10	12	0.3	9	9.7	3.6	—	—	—	—	NK 10/12	RNAF 102012
	IR 71012.5	—	7 -0.008	10	12.5	0.3	9	9.7	3.9	—	—	—	—	—	—
	IR 71015.5	—	7	10	15.5	0.3	9	9.7	4.8	—	—	—	—	—	—
	IR 71016	—	7	10	16	0.3	9	9.7	5	—	—	—	—	NK 10/16	—
8	**IR 81011**	—	8	10	11	0.15	9.2	9.7	2.4	RNA 498	—	—	—	—	—
	IR 81210	—	8	12	10	0.3	10	11	4.8	—	—	—	—	—	—
	IR 81210.5	—	8 0	12	10.5	0.3	10	11	5.1	—	—	—	—	—	—
	IR 81212.5	—	8 -0.008	12	12.5	0.3	10	11	6	—	—	—	—	—	—
	IR 81215.5	—	8	12	15.5	0.3	10	11	7.5	—	—	—	—	—	—
9	**IR 91211**	—	9	12	11	0.3	11	11.5	3.1	RNA 499	—	—	—	—	—
	IR 91212	—	9 0	12	12	0.3	11	11.5	4.5	—	—	—	—	NK 12/12	RNAF 122212
	IR 91216	—	9 -0.008	12	16	0.3	11	11.5	6	—	—	—	—	NK 12/16	—
10	**IR 101312.5**	—	10	13	12.5	0.3	12	12	5.2	—	—	—	—	—	—
	IR 101412	—	10	14	12	0.3	12	13	7	—	—	—	—	—	RNAF 142612
	IR 101412.5	—	10	14	12.5	0.3	12	13	7.2	—	—	—	—	—	—
	IR 101413	—	10	14	13	0.3	12	13	7.5	RNA 4900	—	—	—	—	RNAF 142213
	—	**IRZ 101414**	10	14	14	0.3	12	13	8	RNA 4900UU	—	—	—	—	—
	IR 101416	—	10	14	16	0.3	12	13	9	—	—	—	—	NK 14/16	—
	IR 101416.5	—	10 0	14	16.5	0.3	12	13	9.6	—	—	—	—	—	—
	IR 101420	—	10 -0.008	14	20	0.3	12	13	11.5	—	—	—	—	NK 14/20	RNAFW 142220
	IR 101420.5	—	10	14	20.5	0.3	12	13	11.9	—	—	—	—	—	—
	IR 101510.5	—	10	15	10.5	0.3	12	14	7.9	—	—	—	—	—	—
	IR 101515.5	—	10	15	15.5	0.3	12	14	11.7	—	—	—	—	—	—
	IR 101520.5	—	10	15	20.5	0.3	12	14	15.5	—	—	—	—	—	—
	IR 101525.5	—	10	15	25.5	0.3	12	14	19.3	—	—	—	—	—	—

INNER RINGS

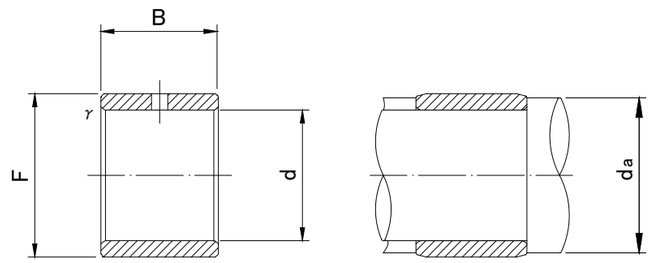






IR





IRZ

IR,IRZ TYPE

Shaft Diameter (mm)	Designation		Dimensions (mm)				Standard mounting dimensions (mm)		Mass	Usable bearing designation					
	IR	IRZ	d	F	B	r s min	da MIN	da MAX	g (approx)	RNA 49	RNA 59	RNA 69	RNA 48	NK	RNAF
12	IR 121512.5	—	12	15	12.5	0.3	14	14	6.1	—	—	—	—	—	—
	IR 121516	—	12	15	16	0.3	14	14	7.8	—	—	—	—	—	—
	IR 121516.5	—	12	15	16.5	0.3	14	14	8.1	—	—	—	—	—	—
	IR 121522.5	—	12	15	22.5	0.3	14	14	11	—	—	—	—	—	—
	IR 121612	—	12	16	12	0.3	14	15	8	—	—	—	—	—	RNAF 162812
	IR 121612.5	—	12	16	12.5	0.3	14	15	8.5	—	—	—	—	—	—
	IR 121613	—	12	16	13	0.3	14	15	8.5	RNA 4901	—	—	—	—	RNAF 162413
	—	IRZ 121614	12	16	14	0.3	14	15	9.6	RNA 4901UU	—	—	—	—	—
	IR 121616	—	12 0 -0.008	16	16	0.3	14	15	10.5	—	—	—	—	NK 16/16	—
	IR 121616.5	—	12	16	16.5	0.3	14	15	11.2	—	—	—	—	—	—
	IR 121620	—	12	16	20	0.3	14	15	13.5	—	—	—	—	NK 16/20	RNAFW 162420
	IR 121620.5	—	12	16	20.5	0.3	14	15	13.9	—	—	—	—	—	—
	IR 121622	—	12	16	22	0.3	14	15	14.5	—	—	RNA 6901	—	—	—
	IR 121622.5	—	12	16	22.5	0.3	14	15	15.2	—	—	—	—	—	—
	—	IRZ 121623	12	16	23	0.3	14	15	15.5	—	—	RNA 6901UU	—	—	—
	IR 121715.5	—	12	17	15.5	0.3	14	16	13.6	—	—	—	—	—	—
	IR 121720.5	—	12	17	20.5	0.3	14	16	18	—	—	—	—	—	—
14	IR 141717	—	14 0 -0.008	17	17	0.3	16	16	9.5	—	—	—	—	—	—
15	IR 151812.5	—	15	18	12.5	0.3	17	17	7.5	—	—	—	—	—	—
	IR 151815.5	—	15	18	15.5	0.3	17	17	9.3	—	—	—	—	—	—
	IR 151816.5	—	15	18	16.5	0.3	17	17	9.9	—	—	—	—	—	—
	IR 151820.5	—	15	18	20.5	0.3	17	17	12.3	—	—	—	—	—	—
	IR 151825.5	—	15	18	25.5	0.3	17	17	15.2	—	—	—	—	—	—
	IR 151916	—	15	19	16	0.3	17	18	12.5	—	—	—	—	NK 19/16	—
	IR 151916.5	—	15	19	16.5	0.3	17	18	13.6	—	—	—	—	—	—
	IR 151920	—	15	19	20	0.3	17	18	16	—	—	—	—	NK 19/20	—
	IR 151920.5	—	15 0	19	20.5	0.3	17	18	12.3	—	—	—	—	—	—
	IR 152012	—	15 -0.008	20	12	0.3	17	19	12	—	—	—	—	—	RNAF 203212
	IR 152013	—	15	20	13	0.3	17	19	13.5	RNA 4902	—	—	—	—	RNAF 202813
	—	IRZ 152014	15	20	14	0.3	17	19	14.5	RNA 4902UU	—	—	—	—	—
	IR 152015.5	—	15	20	15.5	0.3	17	19	16.4	—	—	—	—	—	—
	IR 152018	—	15	20	18	0.3	17	19	19	—	RNA 5902	—	—	—	—
	IR 152020.5	—	15	20	20.5	0.3	17	19	21.5	—	—	—	—	—	—
	IR 152023	—	15	20	23	0.3	17	19	24	—	—	RNA 6902	—	—	—
	—	IRZ 152024	15	20	24	0.3	17	19	25	—	—	RNA 6902UU	—	—	—
	IR 152026	—	15	20	26	0.3	17	19	28	—	—	—	—	—	RNAFW 202826